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Global epidemiology of COVID-19 knowledge, attitude and practice - a systematic review and meta-analysis

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Global epidemiology of COVID-19 knowledge, attitude and practice - a systematic review and meta-analysis

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Global epidemiology of COVID-19 knowledge, attitude and practice - a systematic review and meta-analysis

Objective
To assess the knowledge, attitude and practice (KAP) of the global general population regarding COVID-19.

Design
Systematic review and meta-analysis.

Methods
Medline, Embase, CINAHL and PsycINFO were used to identify articles published between 1 January and 17 December 2020 assessing KAP regarding COVID-19 in the global general population. The quality of eligible studies was assessed. Random effects model was used to obtain the pooled proportion of each component of KAP pertaining to COVID-19. Heterogeneity (I^2) was tested, and subgroup and correlation analyses were performed.

Results
Out of 2,084 records, 50 studies from 26 countries across all continents assessing 85,121 participants' COVID-19 KAP were included in this study. The pooled proportion of KAP was 74% (95% CI: 70–78), 75% (95% CI: 71–80) and 70% (95% CI: 65–75), respectively. Low-income countries, men, people aged 30 and above, and people with 12 years education or less had the lowest practice scores. Practice scores were below 60% in Africa, Europe/Oceania and Americas. Overall heterogeneity was high ($I^2 \geq 98\%$), and publication bias was present (Egger's regression test, $p < 0{\cdot}01$). Very weak correlation was observed between components of KAP.

Conclusions
This study's findings call for community-based awareness programmes to provide a simple, clear and understandable message to reinforce knowledge especially regarding efficacy of each of the preventive measures in low and lower-middle income countries and in Africa, Americas and Europe/Oceania, which will translate into good practice. Targeted intervention for men, people with low education, unemployed people and people aged 30 or older should be recommended. As most of the included studies were cross-sectional online surveys, underprivileged and remote rural people may have been missed out. Additional studies are needed to cover heterogeneous populations.

PROSPERO registration number CRD42020203476.

Strengths and limitations of this study
- This is the first systematic review and meta-analysis that summarises existing KAP pertaining to COVID-19 in the global general population.
- The study identified a large number of studies and participants, covering all continents of the world.
- The methodological design and statistical analyses are sound and robust.
- As most of the included studies were cross-sectional online surveys, underprivileged and remote rural people who do not have internet access may have been missed out.

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Introduction

Emerging infectious diseases (defined as diseases that have 'newly appeared in a population or have existed previously but are rapidly increasing in incidence or geographic range'[1]) have always been a public health threat worldwide. Since late December 2019, the world has been facing Coronavirus Disease 2019 (COVID-19), which is caused by Severe Acute Respiratory Coronavirus 2 (SARS-CoV-2). On 30 January 2020, the World Health Organization (WHO) declared a state of public health emergency. Since then, this emerging and highly infectious disease has spread to 223 countries and territories, leading quickly to a global pandemic, with nearly 120.9 million cases and 2.6 million deaths reported as of 18 March 2021. At time of writing, it is still challenging to contain for most countries around the world.[2] Among infected patients, 0.9%–32% require admission to intensive care units for advanced respiratory support,[3] and the consequent increase in critically ill COVID-19 patients with respiratory failure has overwhelmed intensive care units as well as the invasive mechanical ventilation capacity of healthcare systems in many countries across the globe.

Currently, vaccines are authorised and recommended to prevent COVID-19. Assuming two vaccine doses per person, according to a study published on 3 December 2020,[4] an estimated 15.6 billion doses of COVID-19 vaccines are needed for the 194 WHO member states to achieve a universal COVID-19 vaccination program. Furthermore, 10.3 billion doses are required for targeted and high-risk groups. According to 12 COVID-19 vaccine manufacturers across the world, about 10 billion doses will be available by the end of 2021. Vaccine production, however, is only the first step in a long delivery process that may face immense challenges such as logistics, cold chains and the actual administering of the vaccine.[4] Furthermore, according to WHO, about 50% of the vaccine will be wasted because of temperature control in the supply chains.[5] Therefore, distribution of vaccines, especially in rural and remote communities in many low- to lower-middle-income countries, requires strong international and national supply chains.[4] In addition, the literature shows that vaccine hesitancy may affect the benefits of vaccination campaigns.[6] Thus, until the distribution of vaccine produces herd immunity, perceived-risk-induced behaviours and preventive measures work best to stop the further spread of the disease. This includes behaviours like avoidance of crowded places, washing hands regularly or using hand sanitisers, avoiding frequent touching of the eyes, nose and mouth, cleaning touched surfaces frequently, wearing a mask, social distancing and seeking medical advice upon noticing symptoms like sore throat, fever or shortness of breath.[7, 8] People need to follow these measures voluntarily for the safety of themselves and other people in general. The level to which they follow them voluntarily will depend on their accurate knowledge about the pandemic and their attitude towards it.[9] However, since voluntary action is not enough, most governments across the world have been forced to come up with strict laws and measures that aim to stop the further spread of the disease. Still many countries across the world have not been successful in enforcing these laws. The biggest challenge with people's devotion to these control measures is determined by what they know about the disease, what their attitude is and what practices they undertake to follow their local measures.

Since the emergence of the virus, through the combined efforts of healthcare professionals and government bodies, several measures have been developed to raise awareness, improve knowledge and strengthen preventive practices to control the spread of infection.[10] However, lack of knowledge about mode of transmission and vulnerable populations, as well as lack of attention to preventive measures, may still be widespread; consequently, many countries are facing localised spikes or the localised return of a large number of cases and deaths.

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To the best of our knowledge, no global meta-analysis has been done to pool the overall KAP of COVID-19 of the global general population. Thus, the aim of the current study was to perform a systematic review and meta-analysis in order to provide the current global epidemiology of COVID-19 KAP in the general population.

Methods

This systematic review and meta-analysis was conducted in accordance with 'Preferred Reporting Items for Systematic Reviews and Meta-analyses' (PRISMA)[11] and has been registered on PROSPERO (CRD42020203476). The review process is shown in the flow diagram in Figure 1.

Eligibility criteria

Inclusion criteria: The review included studies that reported any form of quantitative assessment/measurement/evaluation of KAP regarding COVID-19 in the general population in any country or region across the world. There were no restrictions on the age, gender, ethnicity or health status of the participants, the duration for which the study was conducted or the geographic location of the study. Only published articles that had full text available (in English) were included.

Exclusion criteria: Studies were excluded if they were conducted only on some specific group of people, such as healthcare workers, medical students, pregnant women or people with co-morbidities. Short reports, case reports, briefs, letters, editorials and study duplicates were also excluded.

Search strategy, information sources and study selection

With the help of a senior medical librarian, two authors (BS and AS) independently searched the publicly available KAP in COVID-19-related published articles, using databases of MEDLINE, EMBASE, CINAHL and PsycINFO following the PRISMA-P checklist (Supplementary Table 1).[11] This review also included pre-prints, which provide a dynamic update of research papers related to COVID-19. The main keywords for the search strategy included "knowledge", "attitude", "practice" and "COVID-19". Only English-language articles were considered, as a search restriction. EndNote was used to store, organise and manage all the citations.

The same authors (BS and AS) then independently conducted the initial screening of the title and abstract for each article obtained from the literature search. When a title or abstract was not sufficient grounds to reject an article, the full text was obtained and assessed. After the independent review, any disagreement on article inclusion was resolved by discussion or by consulting with the senior author (BB). Reference lists of the articles selected in this meta-analysis were also screened manually for any additional studies.

The preliminary search, which was carried out on 4 September 2020 and updated on 17 December 2020, covered published articles between 1 January 2020 (as the first novel case of COVID-19 was discovered on 31 December 2019) and 17 December 2020. The detailed search strategy is given in the Supplementary Table 2.

Study outcomes

This meta-analysis addressed three main outcomes of overall KAP pertaining to COVID-19 in the general population across the world.

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Knowledge: Level of knowledge or awareness regarding symptoms, mode of transmission, people at risk, incubation and isolation periods, virus fatality, treatments and prevention of COVID-19 infection.
Attitude: Attitude towards controlling or managing of COVID-19.
Practice: Practices such as hand hygiene, wearing face masks, social distancing, avoidance of crowded places/social events and isolation/quarantine to prevent the spread of COVID-19.

The secondary outcomes included the estimated effect by age (under 30 years of age vs 30 or older), gender (male vs female), employment status (unemployed vs employed vs retired/student), income level of the country where the study was conducted (low-income vs lower-middle income vs upper-middle income vs high income), education level (up to 12 years vs over 12 years) and continent.

Data extraction

Using a data extraction template (Excel spreadsheet), two reviewers (BS and AS) independently extracted key information regarding identification of the study (authors, publication month, country where the studies were conducted), methodology (study setting, study design, study population, sample size, data collection tool, recruitment method, recruitment period), participants' demographics (gender, age, education, current employment status, income, area of residence) and main study findings (prevalence of correct knowledge/mean knowledge score of participants regarding symptoms, mode of transmission and prevention of COVID-19, prevalence of positive attitudes of participants towards controlling or managing of COVID-19 and prevalence of good practices such as social distancing, hand hygiene, wearing face masks and (where relevant) quarantine of participants to prevent COVID-19). The two data files were then cross-checked by a third reviewer (OB). Any disagreements were resolved by a consensus meeting under the supervision of a senior author (BB). Missing data, where required, was sought from the study authors.

Quality assessment and risk of bias in individual studies

The quality of the included studies was assessed by two independent reviewers (BS and AS) using the National Heart Lung and Blood Institute (NHLBI) quality assessment tool for observational cohort and cross-sectional studies.[12] The tool uses 14 criteria to assess internal validity and risk of bias. Each criterion was rated as 'yes', 'no', 'cannot determine', 'not applicable' or 'not reported' (Supplementary Table 3). An overall quality of the study was then rated as good, fair or poor. Any discrepancies in the rating between the two reviewers were resolved by a third author (OB).

Data Analysis

Quantitative data (e.g., percentage of KAP regarding COVID-19, and by age, gender, education level, current employment status of the participants) extracted from each study was analysed by the author AA and was then cross-checked by the senior author BB; any discrepancies were resolved. Statistical software package Stata-Version 16 (Statacorp, USA) was used for statistical analyses.

Means (standard deviations) or medians (25th and 75th percentiles), where appropriate, were extracted and then reported for numerical data. Percentage was extracted and then reported for each category of KAP, age group, gender, educational level and employment status of the study participants. The pooled proportion of COVID-19 KAP was calculated as a data synthesis of primary outcomes and was presented in forest plots. This analysis was performed using the random-effects model, as this method demonstrates better properties in the presence of

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heterogeneity (if any), accounting for both within-study and between-study variances.[13] Heterogeneity among the included studies was tested using the χ^2 test on Cochran's Q statistic, which was calculated by means of H and I^2 indices. The I^2 index represents the percentage of total heterogeneity across studies based on true between-study differences rather than on chance. I^2 values of 0–25% show low heterogeneity, 26–75% show moderate heterogeneity and 76–100% show substantial heterogeneity.[14] In order to identify the possible sources of substantial heterogeneity, subgroup analysis was conducted for each of the components of KAP by age group, gender, education level, employment status, income level of the country, continent and China vs other countries.[14] The income level of each country was classified as low-income, lower-middle income, upper-middle income or high income, according to the World Bank.[15] Egger's regression test was used to examine for publication bias, and the symmetry of the funnel plots was evaluated.[16] Sensitivity analyses were performed using quality score categories of the included studies. Furthermore, correlation analysis among the components of KAP was also performed.

Patient and public involvement

There was no patient or public involvement in this systematic review of published literatures.

Results

A total of 2,084 articles, published between 1 January 2020 and 17 December 2020, were retrieved from the four databases and manual searches. After removing duplicate records and screening by titles and abstracts, 82 articles were included for full-text reading. Of these, 32 articles were excluded by the exclusion criteria. Finally, 50 studies from 26 countries or territories, reporting 85,121 participants' KAP towards COVID-19, were included in the qualitative and quantitative analysis.[9, 17-65] Of them, 40 were cross-sectional online surveys, five were community-based, four were both community-based and online surveys and one was a hospital-based cross-sectional study. Based on the National Heart Lung and Blood Institute (NHLBI) quality assessment tool, 12 studies were of good quality, 23 were of fair quality and the remaining 15 were of poor quality. Supplementary Table 4 summarises the study features and the characteristics of the participants. The age of the study participants ranged between 16 and 70 years, and 42·3% were male. In the context of participants' education, 31% had up to secondary-level education, and 69% had university-level education. About 59·5% of participants were employed, 15·7% were unemployed, 16·4% were students and 5·9% were retired. In terms of country income level, 26% of the studies (n = 29,789) were from high-income countries, 26% (n = 26,360) were from upper-middle-income countries, 38% (n = 25,391) were from lower-middle-income countries and 10% (n = 3,715) were from low-income countries.

Figure 2 presents the pooled proportion of KAP of COVID-19 in the study participants. The results showed that 74% had good knowledge (95% CI 70–78%, $p < 0{\cdot}001$, $I^2 = 99{\cdot}40\%$), 75% had positive attitudes (95% CI 71–80%, $p < 0{\cdot}001$, $I^2 = 99{\cdot}70\%$) and 70% had good practice (95% CI 65–75%, $p < 0{\cdot}001$, $I^2 = 99{\cdot}75\%$). High heterogeneity was observed ($I^2 > 99\%$), with the presence of publication bias (Egger's regression test, $p < 0{\cdot}01$). Correlation analysis among the components of KAP was also performed and is presented in Supplementary Figure 1. Very weak and insignificant correlation of knowledge with both attitude and practice was observed. However, attitude had a moderately weak positive correlation with practice ($r = 0{\cdot}314$, $p = 0{\cdot}06$). Furthermore, practice measures pertaining to COVID-19 did not correlate with time since the emergence of this pandemic.

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Supplementary Table 5 presents the subgroup analysis results by age group, gender, education level, employment status, country income level, and continents. Analysis by age group showed that respondents aged below 30 years had similar knowledge (78%, 95% CI 70–85%) but relatively higher practice (87%, 95% CI 81–94%) compared to those aged 30 and above (77%, 95% CI 69–85 and 84%, 95% CI 75–93%, respectively). However, higher positive attitude was observed among participants aged 30 years and above (76%, 95% CI 61–91%) than those under 30 (69%, 95% CI 48–90). There were no significant differences between male and female participants regarding knowledge, attitude and practice, however practice scores were slightly lower among male participants (76%, 95% CI 69−82) than females (78%, 95% CI 72–84%). Participants with 12 years of education or less had lower knowledge and practice than, but similar attitudes to, those with above 12 years of education. Unemployed participants had lower knowledge (70%, 95% CI: 60–79%) and attitude (74%, 95% CI: 62–86%) but higher practice (79%, 95% CI: 70–87%) than employed participants (74%, 95% CI: 67–82%, 79%, 95% CI: 72–87% and 75%, 95% CI: 64–86%). Knowledge score ranged between 71% (95% CI 65–77%) for lower-middle income countries and 80% (95% CI 74–87%) for low-income countries. There was no progressive increase or decrease in attitude and practice scores by income level. However, lower-middle income countries had the lowest attitude score (70%, 95% CI 63–78%) and that was the highest for upper-middle income countries (79%, 95% CI 71–88%). In the context of practicing scores, low-income countries had the lowest (49%, 95% CI 28–71%), and the highest came from the lower-middle income countries (75%, 95% CI 69–81%). Analysis by continent showed that Americas had the highest knowledge score (78%, 95% CI 74−82) and that was the lowest in Europe/Oceania (61%, 95% CI 54−67). Asia had the highest attitude and practice scores (78%, 95% CI 71−84 and 76%, 95% CI 69−82 respectively). However, practice scores were below 60% in Africa, Europe/Oceania and Americas. Heterogeneity was high ($I^2 > 98\%$) across all subgroup analyses, and publication bias was present in most of the studies.

The analyses comparing KAP between studies from China versus other counties are illustrated in Supplementary Figure 2. The KAP scores across four studies from China (n = 10,482) reported an overall higher KAP of 81% (95% CI 65–96%), 95% (95% CI 92–97%) and 87% (95% CI 73–100%) as compared to studies from other countries of 73% (95% CI 70–77%), 73% (95% CI 68–77%) and 68% (95% CI 63–74%), respectively. Sensitivity analyses were performed by quality of the studies, as presented in Supplementary Figure 3; however, high heterogeneity and presence of publication bias still existed.

Further analysis was also done on some important individual questions regarding knowledge, attitude and practices towards COVID-19, as illustrated in Supplementary Table 6. Of all participants, 83% had correct knowledge on both ‘main symptoms’ (reported by 37 studies) and ‘population at risk’ (reported by 30 studies) of COVID-19, while 82% (reported by 36 studies) and 74% (reported by 24 studies) of respondents were aware of ‘route of transmission’ and spread of infection by asymptomatic patients, respectively. Knowledge regarding avoidance of crowded places as a preventive measure was reported by 15 studies, with overall correct knowledge of 95%. Correct knowledge about ‘isolation and treatment of patients’ (reported by 18 studies) and ‘quarantine or incubation period’ (reported by 29 studies) of COVID-19 was observed among 91% and 87% of the study participants respectively. However, only 64% of the participants, reported in 21 studies, had correct knowledge about ‘wearing a face mask’ as a preventive measure.

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Positive attitude towards 'successful control of COVID-19' (reported by 9 studies) and 'social distance' (reported by 5 studies) was reported by 73% and 86% of the study participants respectively.

Practicing hand hygiene through 'proper hand washing or using hand sanitiser' was reported by 83% of the participants in 26 studies; 73% of the study participants (reported by 19 studies) maintained social distance, and 75% (reported by 25 studies) avoided crowded places or social events to prevent the spread of infection. Unexpectedly, only 67% of the participants (reported by 34 studies) were practicing the wearing of face masks while going out. High heterogeneity was observed in all these subgroup analyses, and presence of publication bias was observed in most.

Discussion

This is a large international systematic review and meta-analysis that has examined the knowledge, attitude and practice of the general population on the novel disease COVID-19. Data was compiled from 85,121 participants from 50 studies of 26 countries, covering all continents. The pooled score of KAP was moderate. The overall practice of preventive measures was low in the low-income countries, in Africa, Europe/Oceania and Americas, and among men, employed people and people aged 30 and above. There was a weak correlation between the components of KAP.

The overall knowledge, attitude and practice scores estimated in this review were 74%, 75% 70% respectively. A study from H1N1 influenza A reported a lower knowledge (60%) and similar attitude (74·5%), but higher practice (77·3%) scores, compared to current study.[66] Studies from Severe Acute Respiratory Syndrome (SARS-CoV) in 2003 reported a knowledge score of 52·4–67.4% on symptoms among the general population compared to 83% in our study.[67] Furthermore, a study from Saudi Arabia on Middle East Respiratory Syndrome (MERS) reported study participants' fair knowledge on symptoms (79·8%) and transmissions (90·6%), and their practice on 'washing hands frequently' (94%) and 'using face masks' (74·9%),[68] which were higher than those in the current study. This could be due to high risk perception and their longer experience with MERS (as the study was conducted two years after the first appearance of MERS-CoV).

Subgroup analysis showed that overall knowledge score was low in the lower-middle income countries, in Europe/Oceania and among participants with 12 years or less education, and attitude score was low in Africa, Americas and lower-middle income countries. Low practice score was observed in Africa, Americas, Europe/Oceania, the low income countries, and among males, employed people, aged 30 years or older, and participants with 12 years of education or less. A meta-analysis conducted among general population to see 'the association between gender and protective behaviours in response to respiratory epidemics and pandemics' revealed that female are about 50% more likely than male to adopt or practice preventive behaviours (e.g., hand washing, face mask use, avoidance of public transport).[69] This could be due to females' innate fear for their children and family which drive them to adopt the preventive behaviours or males' earning role in the family which make them to take risks of going out. Education attainment may also have a direct effect on knowledge and consequently on practices as people with less education are less likely to adopt preventive measures. Population density, living conditions, livelihood and affordability could be the reasons behind low practices in the low income countries. Moreover, due to existing health inequalities in low-income and lower-middle-income countries and in socially disadvantaged groups, pandemics generally have their greatest effects on these populations.[4] During the 1918 "Spanish" influenza

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pandemic and in the 2009 H1N1 influenza pandemic, racial minority/minority groups were most affected.[70, 71] However, literature supports that during COVID-19 pandemic, more cases and deaths have been reported from high-income and upper-middle-income countries.[4] These differences may be related to timing of transmission, countries' response patterns, availability and accuracy of tests, socio-demographic condition, countries' environmental factors and connectivity and trade patterns.

Further subgroup analysis showed that overall knowledge, attitude and practice scores were higher (81%, 95% and 87%) in the studies from China, the early COVID-19 epicentre, compared to studies from other countries (73%, 73% and 68%, respectively). This could be due to highly educated study participants in the study from Wuhan (over 80% had associate degrees or higher), overwhelming news by the media and public health authorities, their experience with previous severe acute respiratory syndrome outbreak, and strict actions taken by the local government to control the infection immediately after the outbreak.

In this current study, the overall score of knowledge regarding individual questions about main symptoms, population at risk, route of transmission of COVID-19, isolation and treatment of infected people, and quarantine time was higher than 80%; however, overall, 74% and 64% of respondents had knowledge of asymptomatic patients and the use of face masks as preventive measures, respectively. There were significant differences between knowledge and actual practices, and attitude and practices – specifically, avoidance of crowded places and maintaining social distance. Though 95% of participants had the knowledge of avoidance of crowded places to prevent the infection, only 75% of them were practicing it. On the other hand, 86% of participants had positive attitude to maintain social distance, only 73% of them were practicing it. This difference could be not only due to lack of understanding of the preventive measures, the rationales for their use and their impact on health, but also due to population density and livelihood. Studies from H1N1 influenza A reported correct knowledge on symptoms (68·1–76%), route of transmission (56·3–75·6%), use of face mask (36·6–70·4%), and practice of frequent hand washing (54·6–56·9%) and avoidance of crowded places (42·9–52·6%), which were lower than those in this current study.[72-74] This could be due to low education level (nearly 50% of study participants had below primary/middle school education), lack of perceived risk of spread of the disease and lack of perceived efficacy of various preventive measures among the H1N1 study participants.

A correlation analysis conducted in this study showed a positive but very weak relationship between components of KAP. Literature supports correlation between knowledge, attitude and practice in relation to health issues, which contradicts with this study findings.[75] This may be due to lack of understanding of the severity of the disease or true risk, low perceived risk of acquiring infection and perceived efficacy of the preventive measures. The practice of COVID-19 prevention measures did not increase since the emergence of this pandemic in early 2020, which may be partly reflected from the second and third spikes of COVID-19 spread in many countries across the world.

A wide heterogeneity was observed in the reported prevalence of KAP, ranging between 36–95%, 42–96% and 27–97%, respectively. This was partly explained by the socio-demographic and cultural differences and political influence, and partly by the differences in the questionnaire contents, measurement and scoring systems. The majority of the studies were online surveys, and literature supports that gender, age and education influence online survey behaviour.[21] Internet and social media are today's important online health information resources, hence their users may have gained more information than underprivileged

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population. Differences in the time and actions taken by the government, trust in the governing institutions and experience in management of previous outbreaks may have influenced the attitude of the participants.[21, 28, 40, 65] Mixed message about the use of face masks in public places, scarcity and affordability of face masks, cultural norms to shake hands and going to social, family or religious gatherings, continuity of water supply and hand washing facilities and living conditions may have contributed to adopt the preventive practices against COVID-19 infection.[27, 41, 47, 76]

This systematic review has several strengths and limitations. First, because of the challenges in conducting research during the pandemic, most of the studies were online surveys and covered major cities of the respected countries. Therefore, vulnerable population particularly income-poor people and urban slum dwellers, and people living in remote rural areas who did not have internet access may have missed out. Second, heterogeneity was high in the evaluated outcomes. Although subgroup analyses have been performed, the sources of heterogeneity could not be identified. Third, articles published in other than English were excluded, which may have resulted in the exclusion of some studies. Despite these limitations, to our knowledge, this is the first systematic review and meta-analysis on KAP of COVID-19 among global general population. Furthermore, the study identified a large number of studies and participants, covering all continents of the world, and provided meaningful data on the KAP of COVID-19 in the general population.

Conclusion

KAP pertaining to COVID-19 in the general population in this study seemed moderate with wider ranges. Low practice score was observed in Africa, Americas and Europe/Oceania, in low income countries, and among men, employed, aged 30 years or older, and participants with 12 years of education or less. This data suggest targeted group intervention to reinforce knowledge giving attention to efficacy of each of the preventive measures, which will translate into good practice. Additional studies covering heterogeneous population are also needed.

Figure legend

Figure 1: PRISMA 2009 Flow Diagram

Figure 2: Pooled proportion of knowledge, attitude and practice of COVID-19

Contributors

BB, AA and BS conceived the study and together with AS and OB developed the protocol. BS and AS did the literature search and selected the studies. BS and AS extracted the data, and OB verified the data. AA and BB synthesised the data. BS drafted the initial report with input from AS and BB. BS, AS, OB, AA and BB critically revised successive drafts of the paper. All authors approved the final version of the report.

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Declaration of interests

We declare no competing interests.

Patient consent for publication

Not required.

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Ethics approval

Ethics approval was not required because the analyses in this review are from data that already publicly available in published articles.

Data availability statement

All data generated or analysed during this study are included in this article and its supplementary information files.

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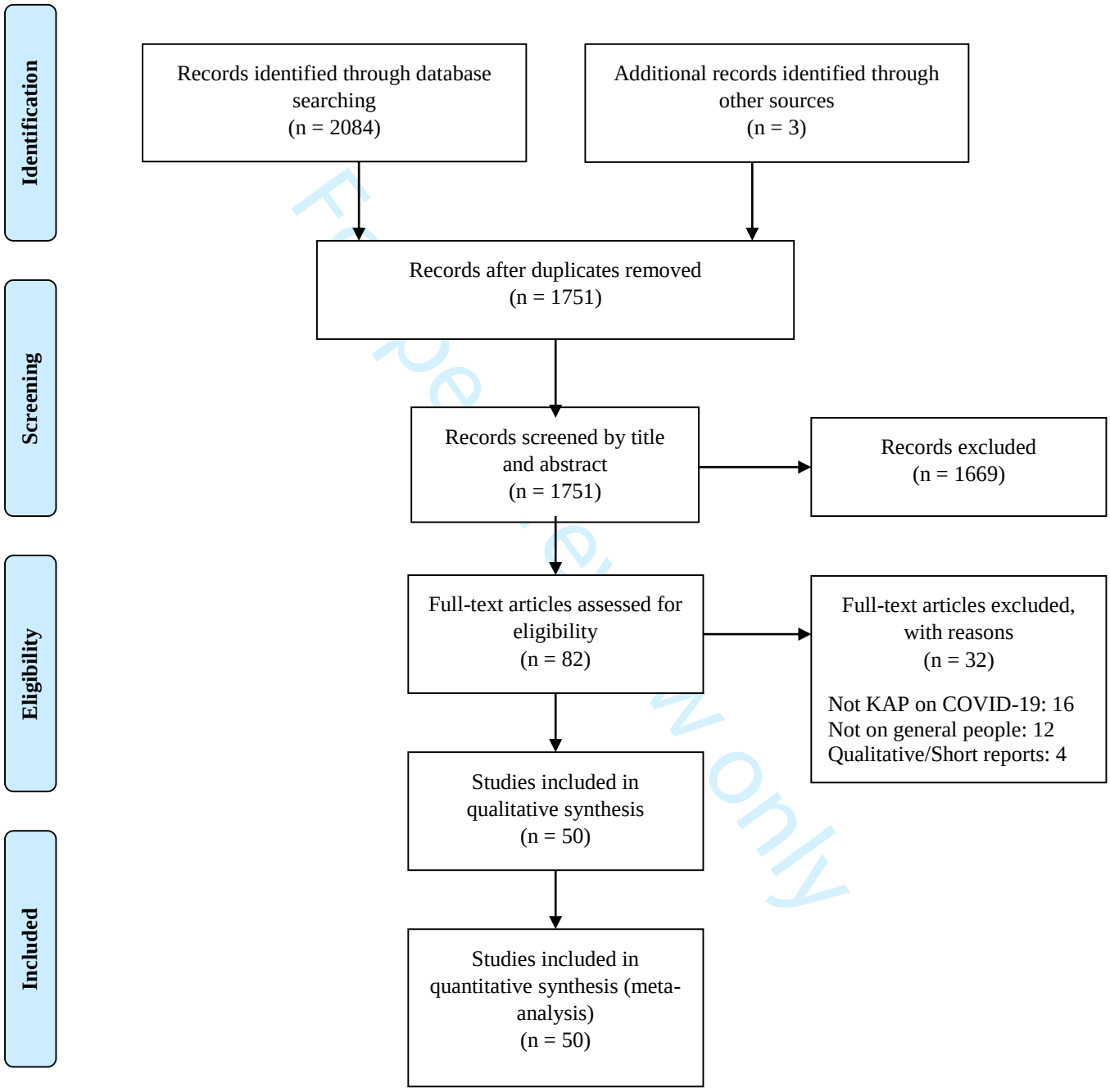
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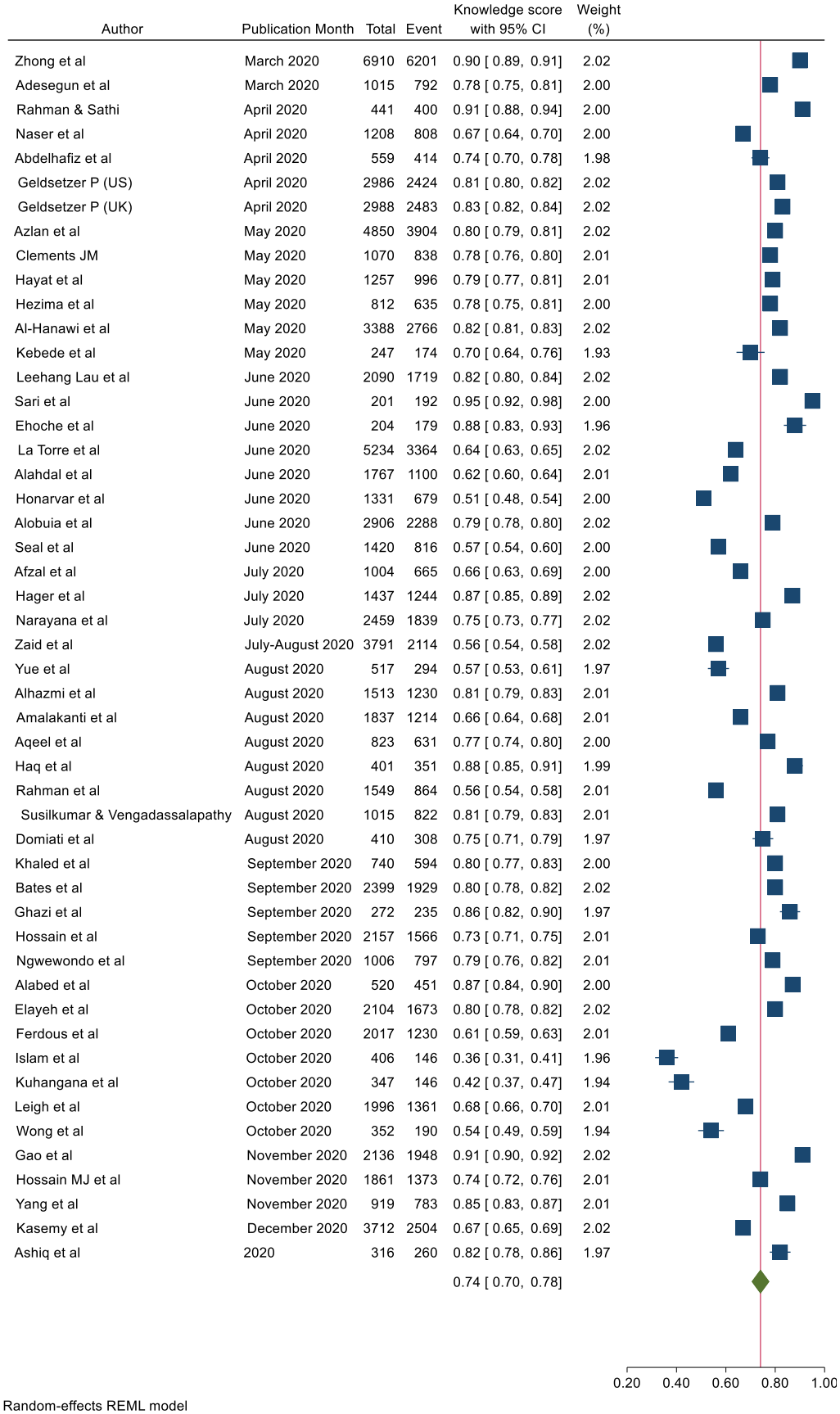
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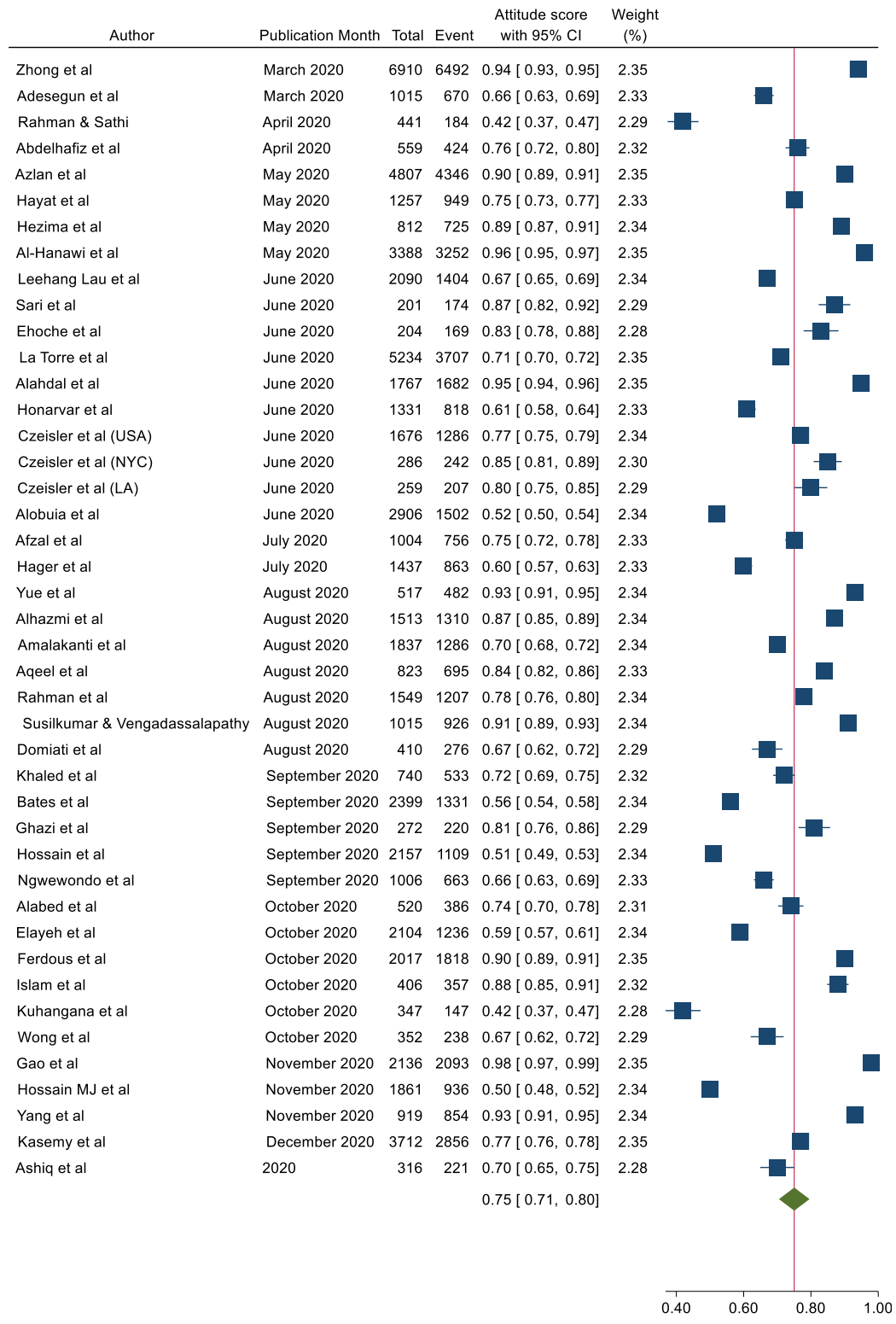
Figure 1: PRISMA 2009 Flow Diagram

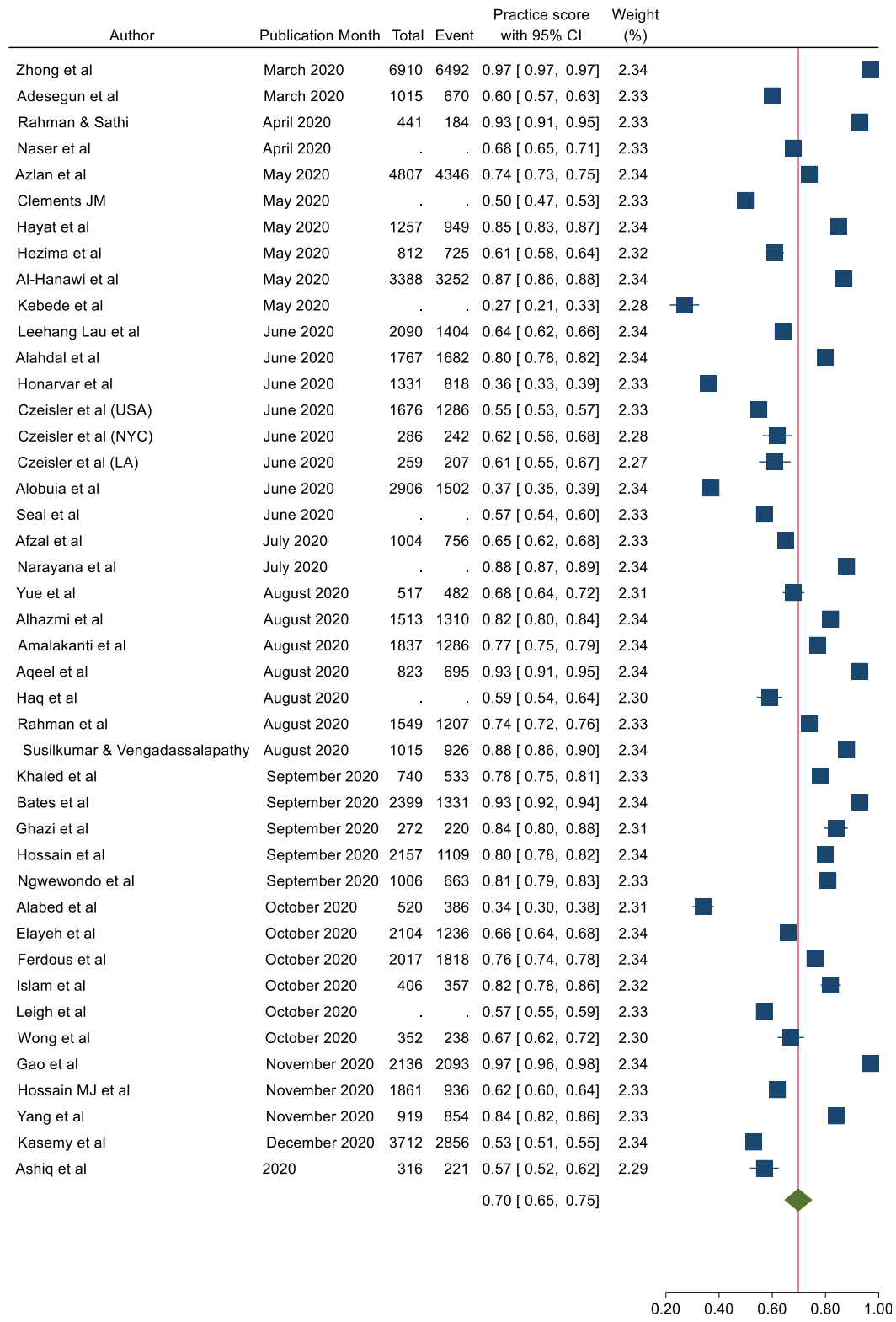


KAP: Knowledge, attitude and practice

Figure 2: Pooled proportion of knowledge, attitude and practice of COVID-19




Author
Publication Month
Total
Event
Attitude score with 95% CI
Weight (%)
Zhong et al March 2020 6910 6492 0.94 [0.93, 0.95] 2.35
Adesegun et al March 2020 1015 670 0.66 [0.63, 0.69] 2.33
Rahman & Sathi April 2020 441 184 0.42 [0.37, 0.47] 2.29
Abdelhafiz et al April 2020 559 424 0.76 [0.72, 0.80] 2.32
Azlan et al May 2020 4807 4346 0.90 [0.89, 0.91] 2.35
Hayat et al May 2020 1257 949 0.75 [0.73, 0.77] 2.33
Hezima et al May 2020 812 725 0.89 [0.87, 0.91] 2.34
Al-Hanawi et al May 2020 3388 3252 0.96 [0.95, 0.97] 2.35
Leehang Lau et al June 2020 2090 1404 0.67 [0.65, 0.69] 2.34
Sari et al June 2020 201 174 0.87 [0.82, 0.92] 2.29
Ehoche et al June 2020 204 169 0.83 [0.78, 0.88] 2.28
La Torre et al June 2020 5234 3707 0.71 [0.70, 0.72] 2.35
Alahdal et al June 2020 1767 1682 0.95 [0.94, 0.96] 2.35
Honarvar et al June 2020 1331 818 0.61 [0.58, 0.64] 2.33
Czeisler et al (USA) June 2020 1676 1286 0.77 [0.75, 0.79] 2.34
Czeisler et al (NYC) June 2020 286 242 0.85 [0.81, 0.89] 2.30
Czeisler et al (LA) June 2020 259 207 0.80 [0.75, 0.85] 2.29
Alobuia et al June 2020 2906 1502 0.52 [0.50, 0.54] 2.34
Afzal et al July 2020 1004 756 0.75 [0.72, 0.78] 2.33
Hager et al July 2020 1437 863 0.60 [0.57, 0.63] 2.33
Yue et al August 2020 517 482 0.93 [0.91, 0.95] 2.34
Alhazmi et al August 2020 1513 1310 0.87 [0.85, 0.89] 2.34
Amalakanti et al August 2020 1837 1286 0.70 [0.68, 0.72] 2.34
Aqeel et al August 2020 823 695 0.84 [0.82, 0.86] 2.33
Rahman et al August 2020 1549 1207 0.78 [0.76, 0.80] 2.34
Susilkumar & Vengadassalapathy August 2020 1015 926 0.91 [0.89, 0.93] 2.34
Domiati et al August 2020 410 276 0.67 [0.62, 0.72] 2.29
Khaled et al September 2020 740 533 0.72 [0.69, 0.75] 2.32
Bates et al September 2020 2399 1331 0.56 [0.54, 0.58] 2.34
Ghazi et al September 2020 272 220 0.81 [0.76, 0.86] 2.29
Hossain et al September 2020 2157 1109 0.51 [0.49, 0.53] 2.34
Ngwewondo et al September 2020 1006 663 0.66 [0.63, 0.69] 2.33
Alabed et al October 2020 520 386 0.74 [0.70, 0.78] 2.31
Elayeh et al October 2020 2104 1236 0.59 [0.57, 0.61] 2.34
Ferdous et al October 2020 2017 1818 0.90 [0.89, 0.91] 2.35
Islam et al October 2020 406 357 0.88 [0.85, 0.91] 2.32
Kuhangana et al October 2020 347 147 0.42 [0.37, 0.47] 2.28
Wong et al October 2020 352 238 0.67 [0.62, 0.72] 2.29
Gao et al November 2020 2136 2093 0.98 [0.97, 0.99] 2.35
Hossain MJ et al November 2020 1861 936 0.50 [0.48, 0.52] 2.34
Yang et al November 2020 919 854 0.93 [0.91, 0.95] 2.34
Kasemy et al December 2020 3712 2856 0.77 [0.76, 0.78] 2.35
Ashiq et al 2020 316 221 0.70 [0.65, 0.75] 2.28
0.75 [0.71, 0.80]
0.40
0.60
0.80
1.00
Random-effects REML model


Author
Publication Month
Total
Event
Practice score with 95% CI
Weight (%)
Zhong et al March 2020 6910 6492 0.97 [0.97, 0.97] 2.34
Adesegun et al March 2020 1015 670 0.60 [0.57, 0.63] 2.33
Rahman & Sathi April 2020 441 184 0.93 [0.91, 0.95] 2.33
Naser et al April 2020 . . 0.68 [0.65, 0.71] 2.33
Azlan et al May 2020 4807 4346 0.74 [0.73, 0.75] 2.34
Clements JM May 2020 . . 0.50 [0.47, 0.53] 2.33
Hayat et al May 2020 1257 949 0.85 [0.83, 0.87] 2.34
Hezima et al May 2020 812 725 0.61 [0.58, 0.64] 2.32
Al-Hanawi et al May 2020 3388 3252 0.87 [0.86, 0.88] 2.34
Kebede et al May 2020 . . 0.27 [0.21, 0.33] 2.28
Leehang Lau et al June 2020 2090 1404 0.64 [0.62, 0.66] 2.34
Alahdal et al June 2020 1767 1682 0.80 [0.78, 0.82] 2.34
Honarvar et al June 2020 1331 818 0.36 [0.33, 0.39] 2.33
Czeisler et al (USA) June 2020 1676 1286 0.55 [0.53, 0.57] 2.33
Czeisler et al (NYC) June 2020 286 242 0.62 [0.56, 0.68] 2.28
Czeisler et al (LA) June 2020 259 207 0.61 [0.55, 0.67] 2.27
Alobuia et al June 2020 2906 1502 0.37 [0.35, 0.39] 2.34
Seal et al June 2020 . . 0.57 [0.54, 0.60] 2.33
Afzal et al July 2020 1004 756 0.65 [0.62, 0.68] 2.33
Narayana et al July 2020 . . 0.88 [0.87, 0.89] 2.34
Yue et al August 2020 517 482 0.68 [0.64, 0.72] 2.31
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Ghazi et al September 2020 272 220 0.84 [0.80, 0.88] 2.31
Hossain et al September 2020 2157 1109 0.80 [0.78, 0.82] 2.34
Ngwewondo et al September 2020 1006 663 0.81 [0.79, 0.83] 2.33
Alabed et al October 2020 520 386 0.34 [0.30, 0.38] 2.31
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Kasemy et al December 2020 3712 2856 0.53 [0.51, 0.55] 2.34
Ashiq et al 2020 316 221 0.57 [0.52, 0.62] 2.29
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Random-effects REML model

Supplementary material

Global epidemiology of COVID-19 knowledge, attitude and practice - a systematic review and meta-analysis

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Table 1: PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Page 4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Page 5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Page 5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Page 5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Page 5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Page 5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Page 5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Page 6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Page 6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Page 6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Page 6-7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Page 7
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Page 7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Page 7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Page 7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Page 7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	-
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Page 7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Page 7-8
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Page 8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Page 9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Page 11
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Page 11
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Page 7

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Table 2: Search strategy in MEDLINE

	Search terms
#1	(wuhan and (coronavirus or corona virus)).tw,kf,hw.
#2	(coronavirus* and ("19" or "2019")).tw,kf,hw.
#3	(2019 nCov or nCov 2019 or ncov19 or ncov 19 or novel coronavirus* or novel corona virus* or Severe Acute Respiratory Syndrome Coronavirus 2 or coronavirus disease 2019 or corona-virus disease 2019 or new coronavirus* or new corona-virus* or SARS-Coronavirus-2 or SARS-Coronavirus2 or SARS-Corona-Virus-2 or SARS-corona-virus2 or SARS-like coronavirus*).tw,kf,hw.
#4	(2019-novel CoV or SARS-COV-2 or SARS-COV2 or sarscov2 or sarscov-2 or coronavirus-19 or covid19 or covid-19).tw,kf,hw.
#5	((novel or new or nouveau or pandemic*) adj2 (CoV or covid or ncov or coronavirus or corona-virus)).tw,kf,hw.
#6	#1 OR #2 OR #3 OR #4 OR #5
#7	Knowledge/
#8	Attitude/
#9	Practice/
#10	Health Knowledge, Attitudes, Practice/
#11	#7 OR #8 OR #9 OR #10
#12	#6 AND #11
#13	limit 12 to english language

Table 3: Tool used for quality assessment

	Criteria	Yes	No	Others (CD, CR, NA)*
1	Was the research question or objective in this paper clearly stated?			
2	Was the study population clearly specified and defined?			
3	Was the participation rate of eligible persons at least 50%?			
4	Were all the subjects selected or recruited from the same or similar populations (including the same time period)? Were inclusion and exclusion criteria for being in the study pre-specified and applied uniformly to all participants?			
5	Was a sample size justification, power description, or variance and effect estimates provided?			
6	For the analyses in this paper, were the exposure(s) of interest measured prior to the outcome(s) being measured?			
7	Was the time frame sufficient so that one could reasonably expect to see an association between exposure and outcome if it existed?			
8	For exposures that can vary in amount or level, did the study examine different levels of the exposure as related to the outcome (e.g., categories of exposure or exposure measured as continuous variable)?			
9	Were the exposure measures (independent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?			
10	Was the exposure(s) assessed more than once over time?			
11	Were the outcome measures (dependent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?			
12	Were the outcome assessors blinded to the exposure status of participants?			
13	Was loss to follow-up after baseline 20% or less?			
14	Were key potential confounding variables measured and adjusted statistically for their impact on the relationship between exposure(s) and outcome(s)?			
	Quality rating (good, fair, or poor) (see guidance) Rater no. 1 initials: Rater no. 2 initials: Additional comments (if poor, please state why):			

*CD: cannot determine; NA: not applicable; NR: not reported.

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Table 4: Studies included in the systematic review and meta-analysis

	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
Asia	Azlan et al, May 2020[27]	Malaysia	Cross-sectional online survey	Malaysian above 18 years	4850	34·0	42·1	Good	3904	80·0	4346	90·4	3564	74·1
	Afzal et al, July 2020[19]	Pakistan	Cross-sectional online survey	Pakistani nationals aged 16 years or more	1004	NR*	37·0	Fair	665	66·2	756	75·3	654	65·2
	Hayat et al, May 2020[40]	Pakistan	Cross-sectional online survey	Pakistani residents age >15yrs	1257	16–30+	44·3	Fair	996	79·2	949	75·5	1073	85·3
	Leehang Lau et al, June 2020[51]	Philippines	Community-based Cross-sectional survey	participants of ICM's poverty alleviation program	2224	41·3	7·3	Fair	1719	82·2	1404	67·2	1328	63·5
	Rahman & Sathi, April 2020[56]	Bangladesh	Cross-sectional online survey	Bangladeshi internet users ≥18 years	441	18–30+	68·7	Poor	400	90·7	184	41·7	408	92·6
	Sari et al, June 2020[58]	Indonesia	Cross-sectional online survey	aged 18–60 years	201	35·6	46·3	Fair	192	95·5	174	86·6	NR	NR
	Yue et al, August 2020[63]	Henan, China	Cross-sectional online survey	People aged 15 years or older	517	15–60	46·2	Fair	294	56·9	482	93·3	350	67·7
	Zhong et al, March 2020[65]	China	Cross-sectional online survey	Chinese nationality aged 16 years or more	6910	33·0	34·3	Fair	6201	89·7	6492	93·9	6717	97·2
	Alabed et al, October 2020[20]	Malaysia	Cross-sectional online survey	People living in Malaysia aged 18 years or above	520	36·9	38·5	Poor	451	86·7	386	74·3	178	34·2
	Amalakanti et al, August 2020[24]	Andhra Pradesh, India	Cross-sectional online survey	South Indian aged above 15 years	1837	16–50+	56·5	Poor	1214	66·1	1286	70·0	1414	77·0
	Aqeel et al, August 2020[25]	Delhi, India	Cross-sectional online survey	Residents of Delhi	823	38·2	56·9	Poor	631	76·6	695	84·4	769	93·4
	Ashiq et al, August 2020[26]	Lahore, Pakistan	Cross-sectional online survey	People residing in Lahore	316	16–40+	46·5	Poor	260	82·3	221	69·9	181	57·2
	Ferdous et al, October 2020[34]	Bangladesh	Cross-sectional online survey	Bangladeshi residents aged 12-64 years	2017	24·4	59·8	Poor	1230	60·9	1818	90·1	1534	76·1

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Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
								Event, n	%	Event, n	%	Event, n	%
Gao et al, November 2020[35]	China	Cross-sectional online survey	Chinese residents 18 years and above	2136	33·1	21·9	Poor	1948	91·2	2093	98·0	2068	96·8
Haq et al, August 2020[39]	Punjab, Pakistan	Cross-sectional online survey	People living in Pakistan >15 years	401	NR	52·3	Poor	351	87·6	NR	NR	236	58·7
Hossain et al, September 2020[43]	Bangladesh	Cross-sectional online survey	Bangladeshi residents 13-90 years	2157	33·5	54·1	Fair	1566	72·6	1109	51·4	1716	79·5
Hossain MJ et al, November 2020[44]	Bangladesh	Cross-sectional community-based and online study	Bangladeshi residents	1861	15−50+	64·5	Poor	1373	73·8	936	50·3	1154	62·0
Islam et al, October 2020[45]	Bangladesh	Cross-sectional community-based study	Aged ≥18 years slum dwellers	406	44·9	53·2	Fair	146	35·9	357	87·9	332	81·7
Narayana et al, July 2020[53]	India	Cross-sectional online survey	Indian residents aged above 15 years	2459	24·5	57·9	Good	1839	74·7	NR	NR	2167	88·1
Rahman et al, August 2020[57]	Dhaka, Rangpur and Bogura, Bangladesh	Cross-sectional online and offline study	Adult population living in Bangladesh	1549	18−61	57·9	Poor	864	55·8	1207	77·9	1140	73·5
Susilkumar & Vengadassalapathy, August 2020[60]	India	Cross-sectional online survey	Participants with access to a primary internet connection and who can read and write the English language	1015	20−60	49·3	Poor	822	81·0	926	91·1	892	87·8
Wong et al, October 2020[61]	Hong Kong	Cross-sectional study	Age of 18 years or older South Asian in Hong Kong	352	38·9	40·3	Fair	190	53·8	238	67·4	234	66·5

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	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
	Yang et al, November 2020[62]	China	Cross-sectional online survey	Residents who were aged 18 years and above, understood Chinese	919	18−46+	21·7	Good	783	85·2	854	92·9	776	84·4
Africa	Abdelhafiz et al, April 2020[17]	Egypt	Cross-sectional community-based and online survey	Egyptian adults	559	18−60+	37·7	Fair	414	74·1	424	75·8	NR	NR
	Ehoche et al, June 2020[32]	Nigeria	Cross-sectional online survey	North Central Nigeria	204	15−60	58·8	Poor	179	87·6	169	82·8	NR	NR
	Hezima et al, May 2020[41]	Khartoum, Sudan	Cross-sectional community-based and online survey	Sudanese residents	812	18−55+	54·2	Good	635	78·2	725	89·3	494	60·8
	Hager et al, July 2020[38]	Egypt and Nigeria	Cross-sectional online survey	adults >17 years of all educational levels	1437	18−59+	52·5	Good	1244	86·6	863	60·0	NR	NR
	Kebede et al, May 2020[47]	Jimma town, South-west Ethiopia,	Hospital-based Cross-sectional study	Visitors of medical centre	247	30·5	76·5	Fair	174	70·4	NR	NR	66	26·9
	Adesegun et al, September 2020[18]	Nigeria	Cross-sectional online survey	Literate nigerian who understand english	1015	26·6	45·9	Poor	792	78·0	670	66·0	613	60·4
	Kasemy et al, December 2020[46]	Egypt	Cross-sectional online survey	Egyptians	3712	23·3	47·8	Fair	2504	67·4	2856	76·9	1950	52·5
	Kuhangana et al, October 2020[49]	Democratic Republic of the Congo	Cross-sectional study	adult people (>18 years) frequenting ten public markets	347	37·4	17·0	Poor	146	42·2	147	42·5	NR	NR
	Ngwewondo et al, September 2020[55]	Cameroon	Cross-sectional online survey	Cameroonian residents, aged 18 years or more	1006	33·0	46·9	Fair	797	79·2	663	65·8	812	80·7
Middle	Al-Hanawi et al, May 2020[9]	Saudi Arabia	Cross-sectional online survey	aged 18 years or older living in KSA	3388	18−60+	41·9	Fair	2766	81·6	3252	95·9	2942	86·8
	Alahdal et al, June 2020[21]	Riyadh, Saudi Arabia	Cross-sectional online survey	18-65 years old	1767	18−65	25·0	Good	1100	62·3	1682	95·2	1406	79·6

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	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
East	Honarvar et al, June 2020[42]	Shiraz, Iran	Cross-sectional and population-based study	people aged at least 15 years	1331	36·0	47·3	Good	679	50·9	818	61·5	481	36·2
	Naser et al, April 2020[54]	Jordan, Saudi Arabia and Kuwait	Cross-sectional online survey	aged 18 years who was living either in Jordan, Saudi Arabia or Kuwait	1208	18–50+	32·8	Fair	808	66·9	NR	NR	808	67·6
	Alhazmi et al, August 2020[22]	Saudi Arabia	Cross-sectional online survey	All citizens and residents over 18 years	1513	NR	45·0	Good	1230	81·3	1310	86·6	1239	81·9
	Domiati et al, August 2020[31]	Lebanon	Cross-sectional online survey	Residents of Lebanon	410	NR	41·9	Good	308	75·0	276	67·2	NR	NR
	Elayeh et al, October 2020[33]	Jordan	Cross-sectional online survey	Adult residents of Jordan	2104	18–55+	24·6	Good	1673	79·5	1236	58·7	1385	65·8
	Ghazi et al, September 2020[37]	Iraq	Cross-sectional online survey	Adults living in Iraq	272	36·4	58·1	Good	235	86·2	220	80·7	227	83·5
	Khaled et al, September 2020[48]	Aseer region, Saudi Arabia	Cross-sectional online survey	Residents of the Aseer Region, Saudi Arabia	740	18–70	11·5	Fair	594	80·3	533	71·9	575	77·6
	Zaid et al, July-August 2020[64]	Jordan	Cross-sectional online survey	Jordanian aged 18 years	3791	18–60	26·8	Poor	2114	55·7	NR	NR	NR	NR
North	Alobuia et al, June 2020[23]	USA	Telephone survey	aged 18 year and older	2906	18–65+	48·0	Fair	2288	78·7	1502	51·7	1073	36·9
	Clements JM, May 2020[29]	USA	Cross-sectional online survey	US residents aged 18 years or older	1070	37·1	58·2	Good	838	78·3	NR	NR	536	50·1

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	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
America	Czeisler et al, June 2020[30]	New York (NYC) and Los Angeles (LA), USA	Cross-sectional online survey	adults aged ≥18 years	1676	18−65+	43·9	Fair	NR	NR	USA (1286) NYC (242) LA (207)	USA (76·7) NYC (84·5) LA (80·0)	USA (920) NYC (178) LA (158)	USA (54·9) NYC (62·2) LA (61·0)
	Geldsetzer P, April 2020[36]	US and UK	Cross-sectional online survey	Adults registered with Prolific Academic Pty	US (2986) UK (2988)		US (49·1) UK (48·8)	Fair	US (2424) UK (2483)	US (81·2) UK (83·1)	NR	NR	NR	NR
	Leigh et al, October 2020[52]	Canada	Cross-sectional online survey	Adult Canadian	1996	18−65	45·2	Fair	1361	68·1	NR	NR	1143	57·2
South America	Bates et al, September 2020[28]	Ecuador	Cross-sectional online survey	Ecuadorians aged 18 or greater	2399	18−50+	37·0	Fair	1929	80·4	1331	55·5	2219	92·5
Europe	La Torre et al, June 2020[50]	Italy	Cross-sectional online survey	11-30 years	5234	11−30	36·7	Fair	3364	64·3	3707	70·8	NR	NR
Oceania	Seal et al, June 2020[59]	Australia	Cross-sectional online survey	Australian adults (18 years and older)	1420	18−50+	48·0	Fair	816	57·4	NR	NR	803	56·5

*NR-Not reported

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Table 5: Subgroup analysis by country income level, continent, age group, gender, education and employment status

Outcome	Study characteristics	No of studies	Estimates score (%)	95% CI*	p-value	I^2	Egger test (p-value)
Knowledge	**Income level of the country**						
	Low income countries	5	80	74–87	<0·001	95·51%	
	Lower-middle income countries	19	71	65–77	<0·001	99·28%	
	Upper-middle income countries	13	78	70–86	<0·001	99·64%	
	High income countries	13	72	67–78	<0·001	99·16%	
	Continent						
	Asia	22	75	69–81	<0·001	99·54%	
	Africa	9	74	65–83	<0·001	98·99%	
	Middle East	9	73	65–81	<0·001	99·22%	
	Americas	6	78	74–82	<0·001	97·47%	
	Europe/Oceania	2	61	54–67			
	Age group	7					
	Below 30 years		78	70–85	<0·001	98·50%	0·059
	30 years and above		77	69–86	<0·001	99·19%	0·665
	Gender	25					
	Male		74	69–79	<0·001	98·90%	0·052
	Female		74	68–79	<0·001	99·37%	0·011
	Education level	20					
	Up to 12 years		70	65–75	<0·001	97·14%	0·584
	Above 12 years		77	71–82	<0·001	99·18%	0·086
	Employment status	10					
	Unemployed		70	60–79	<0·001	100%	0·378
	Employed		74	67–82	<0·001	99·48%	<0·001
	Retired/students		71	61–81	<0·001	98·58%	0·292
Attitude	**Income level of the country**						
	Low income countries	4	74	61–88	<0·001	98·87%	
	Lower-middle income countries	17	70	63–78	<0·001	99·46%	
	Upper-middle income countries	12	79	71–88	<0·001	99·81%	
	High income countries	10	78	70–87	<0·001	99·65%	
	Continent						
	Asia	21	78	71–84	<0·001	99·76	
	Africa	8	70	60–80	<0·001	99·15	
	Middle East	8	77	67–88	<0·001	99·68	
	Americas	5	70	57–83	<0·001	99·29	
	Europe/Oceania	1	71	70–72			
	Age group	5					
	Below 30 years		69	48–90	<0·001	99·83%	<0·001
	30 years and above		76	61–91	<0·001	99·76%	0·069
	Gender	18					
	Male		75	67–83	<0·001	99·60%	0·091
	Female		75	67–82	<0·001	99·71%	0·002
	Education level	13					
	Up to 12 years		71	62–80	<0·001	99·19%	<0·001
	Above 12 years		71	62–80	<0·001	99·70%	<0·001
	Employment status	7					
	Unemployed		74	62–86	<0·001	98·75%	0·108
	Employed		79	72–87	<0·001	99·54%	0·045
	Retired/students		79	69–89	<0·001	99·07%	0·039
Practice	**Income level of the country**						
	Low income countries	3	49	28–71	<0·001	99·00%	
	Lower-middle income countries	17	75	69–81	<0·001	99·31%	
	Upper-middle income countries	10	73	59–88	<0·001	99·93%	
	High income countries	13	65	57–73	<0·001	99·33%	
	Continent						
	Asia	22	76	69–82	<0·001	99·74	
	Africa	5	56	40–73	<0·001	99·48	
	Middle East	8	73	61–84	<0·001	99·58	
	Americas	7	59	47–72	<0·001	99·57	
	Europe/Oceania	1	57	54–60			
	Age group	6					
	Below 30 years		87	81–94	<0·001	98·70%	0·128
	30 years and above		84	75–93	<0·001	99·50%	0·834
	Gender	20					
	Male		76	69–82	<0·001	99·36%	0·002
	Female		78	72–84	<0·001	99·62%	0·001
	Education level	7					

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Outcome	Study characteristics	No of studies	Estimates score (%)	95% CI*	p-value	I^2	Egger test (p-value)
	Up to 12 years		76	69–82	<0·001	98·48%	0·003
	Above 12 years		79	70–87	<0·001	99·75%	<0·001
	Employment status	11					
	Unemployed		79	70–87	<0·001	98·76%	0·034
	Employed		75	64–86	<0·001	99·79%	0·001
	Retired/students		75	64–86	<0·001	99·23%	0·059

*CI: Confidence Interval

Table 6: Analysis by some important questions regarding knowledge, attitude and practice on COVID-19

Outcome	Study characteristics	No of studies	Estimates score (%)	95% CI*	p-value	I^2	Egger test (p-value)
Knowledge	Q1.Main symptoms	37	83	78–89	<0·001	99·89%	<0·001
	Q2.Population at risk	30	83	77–88	<0·001	99·86%	0·0002
	Q3.Route of transmission	36	82	76–87	<0·001	99·89%	<0·001
	Q4.Spread of infection by asymptomatic patients	24	74	66–82	<0·001	99·87%	0·0003
	Q5.Face mask as a preventive measure	21	64	53–74	<0·001	99·92%	0·644
	Q6.Avoidance of crowded places as a preventive measure	15	95	92–97	<0·001	99·65%	<0·001
	Q7.Isolation and treatment of patients to prevent infection	18	91	85–97	<0·001	99·94%	0·022
	Q8.Quarantine/Incubation period	29	87	83–91	<0·001	99·85%	0·008
Attitude	Q1.COVID-19 will be successfully controlled	9	73	60–86	<0·001	99·88%	0·003
	Q2.It is important to maintain social distance	5	86	77–95	<0·001	99·59%	0·199
Practice	Q1.Hand hygiene	26	83	77–89	<0·001	99·95%	<0·001
	Q2.Wearing face mask	34	67	58–77	<0·001	99·97%	0·036
	Q3.Social distancing	19	73	64–83	<0·001	99·82%	0·024
	Q4.Avoiding crowded places	25	75	67–83	<0·001	99·88%	0·041

*CI: Confidence Interval

Figure 1: Correlation between components of KAP and study period with practice

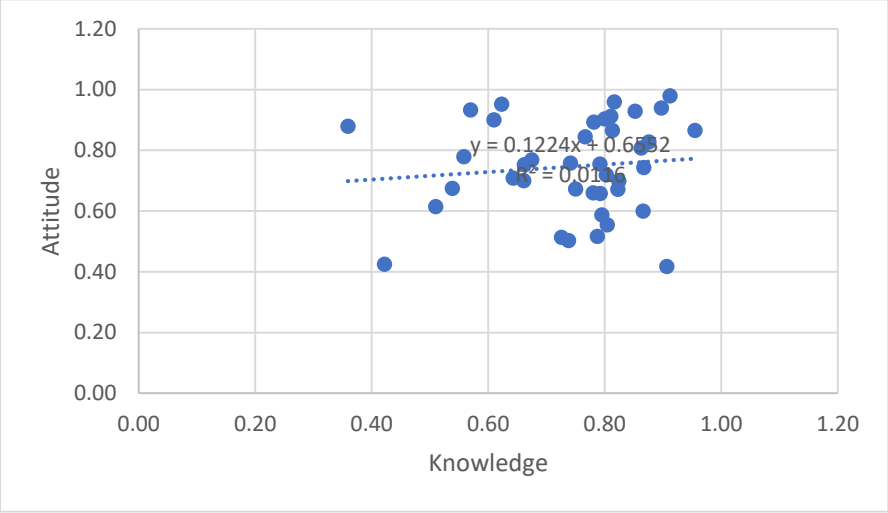


Figure 1a: Correlation between knowledge and attitude

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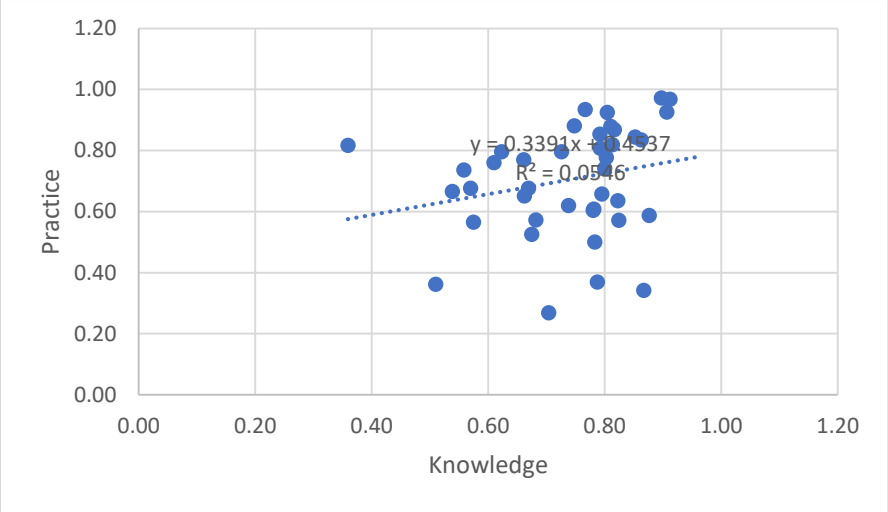


Figure 1b: Correlation between knowledge and practice

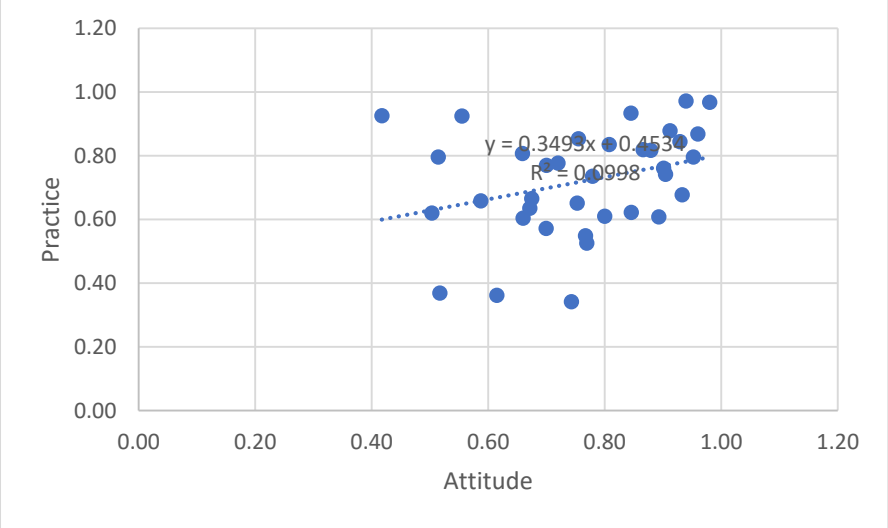


Figure 1c: Correlation between attitude and practice

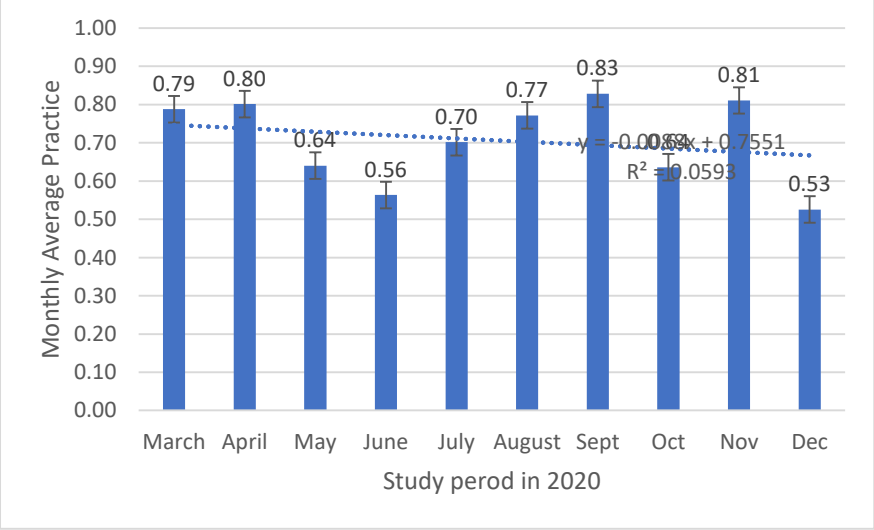
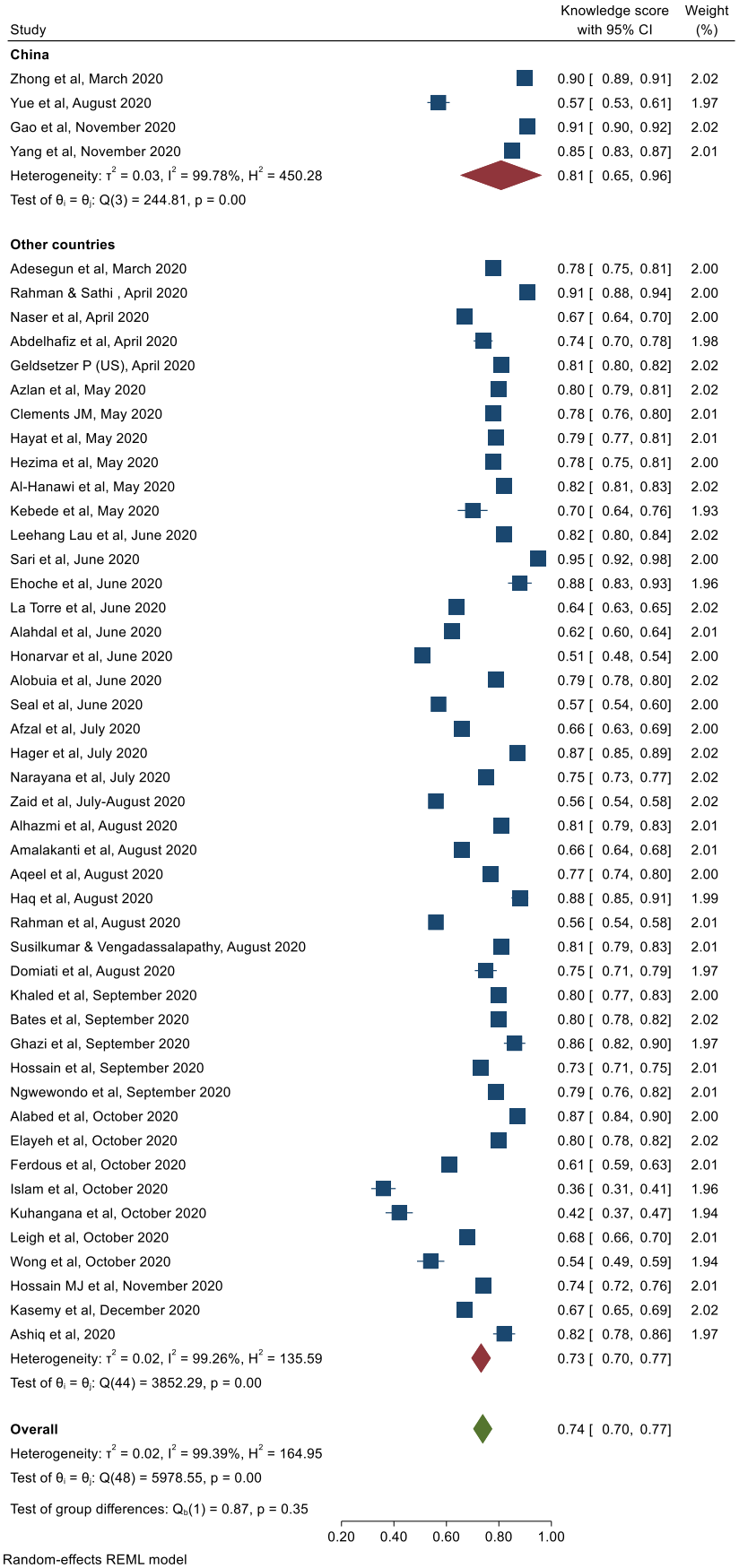


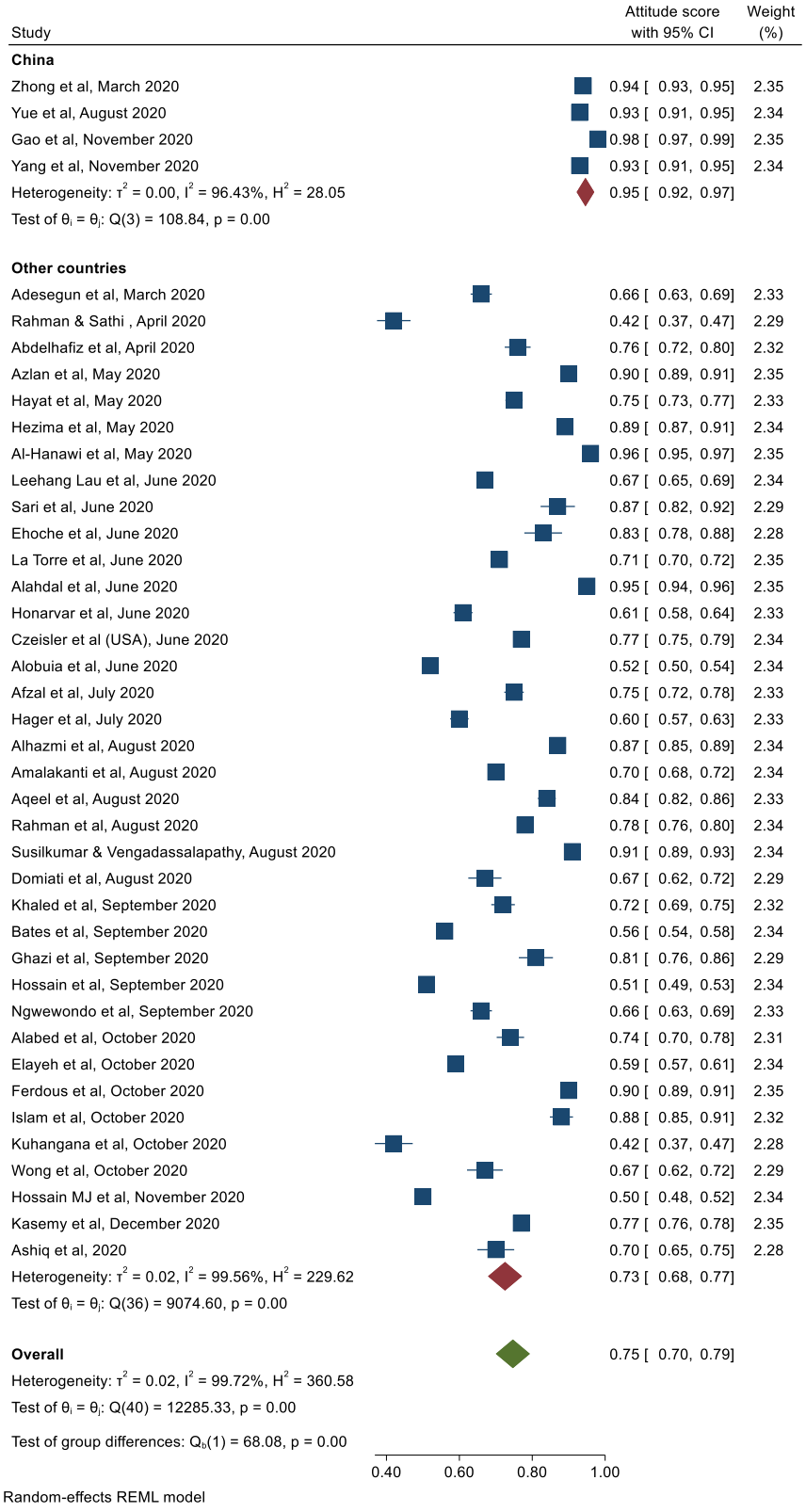
Figure 1d: Correlation between study period and practice

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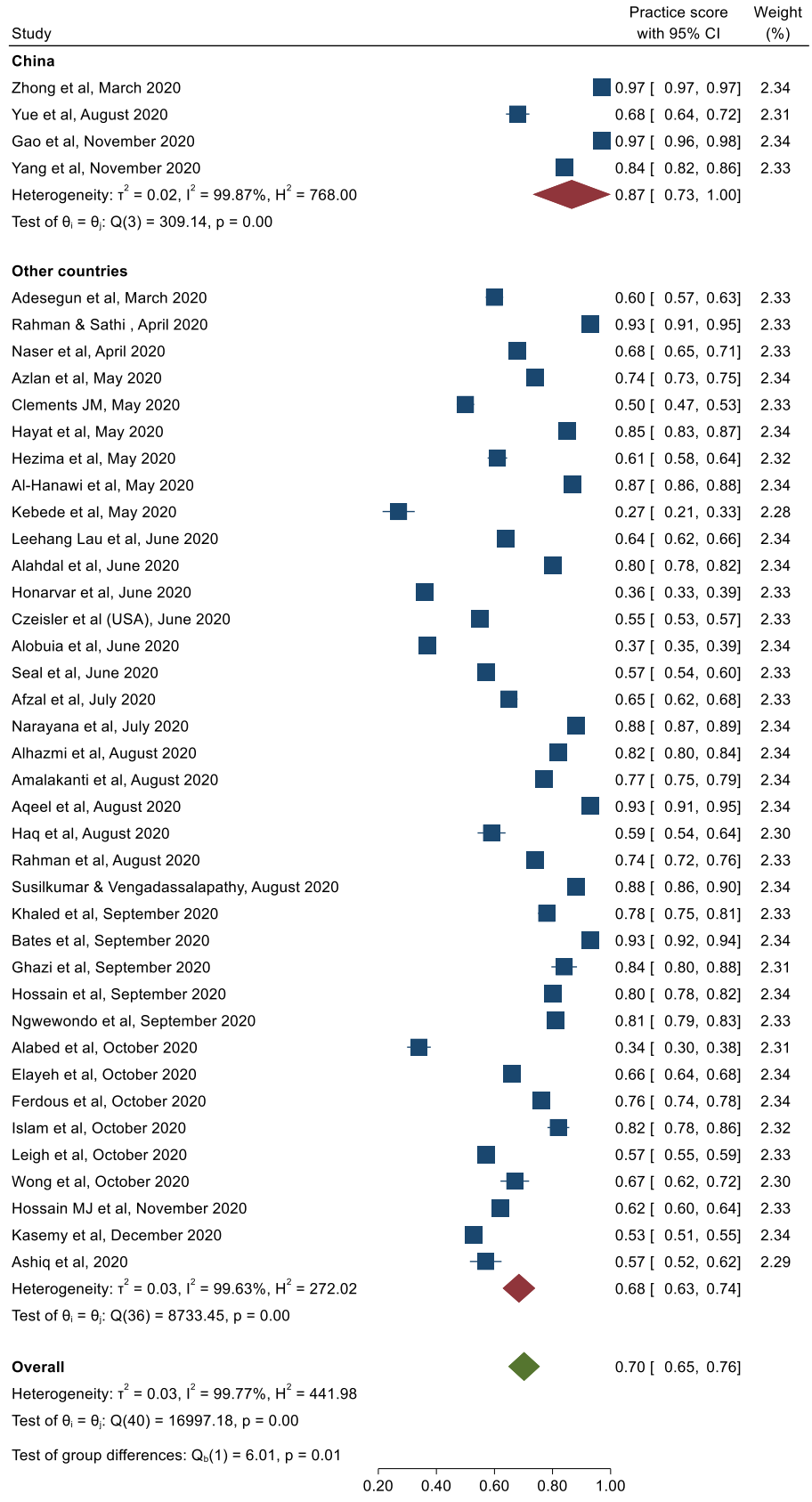
Figure 2: China vs Other countries



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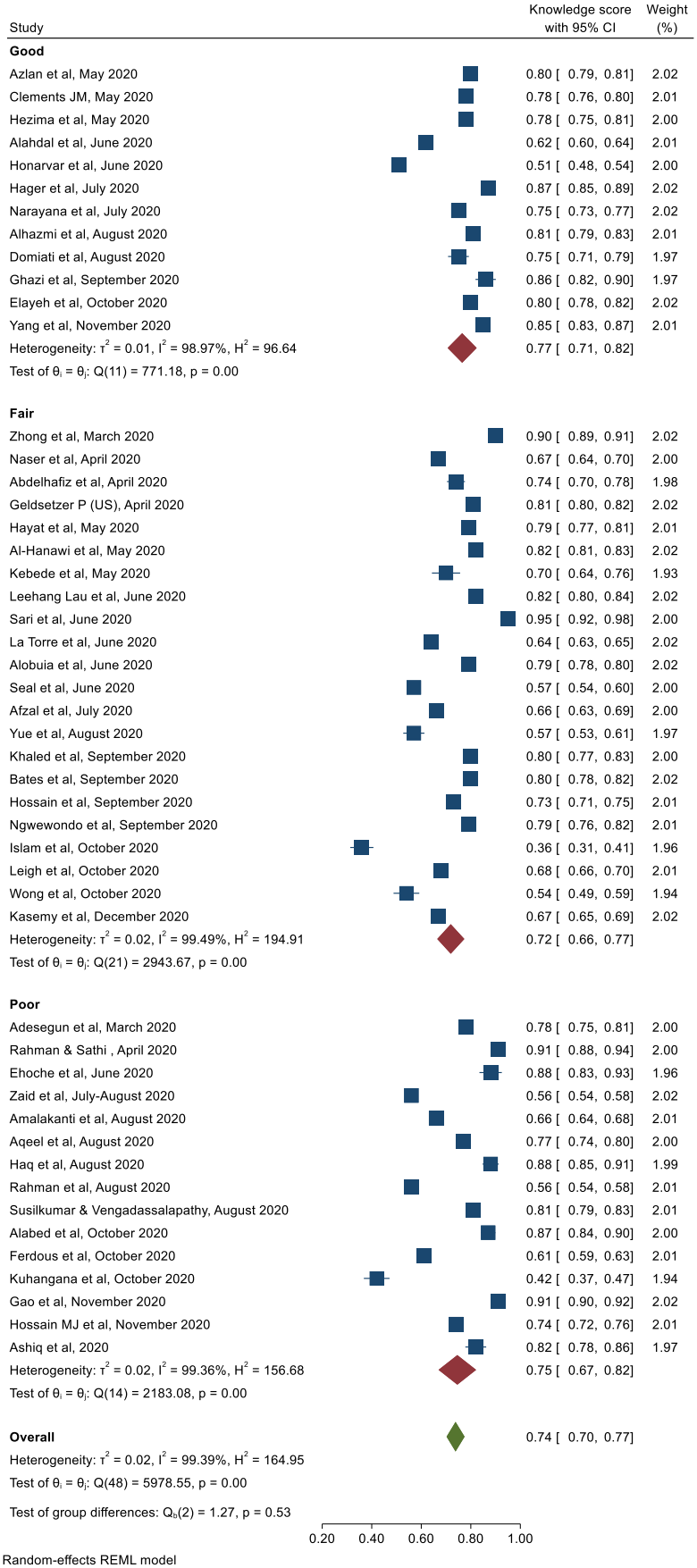


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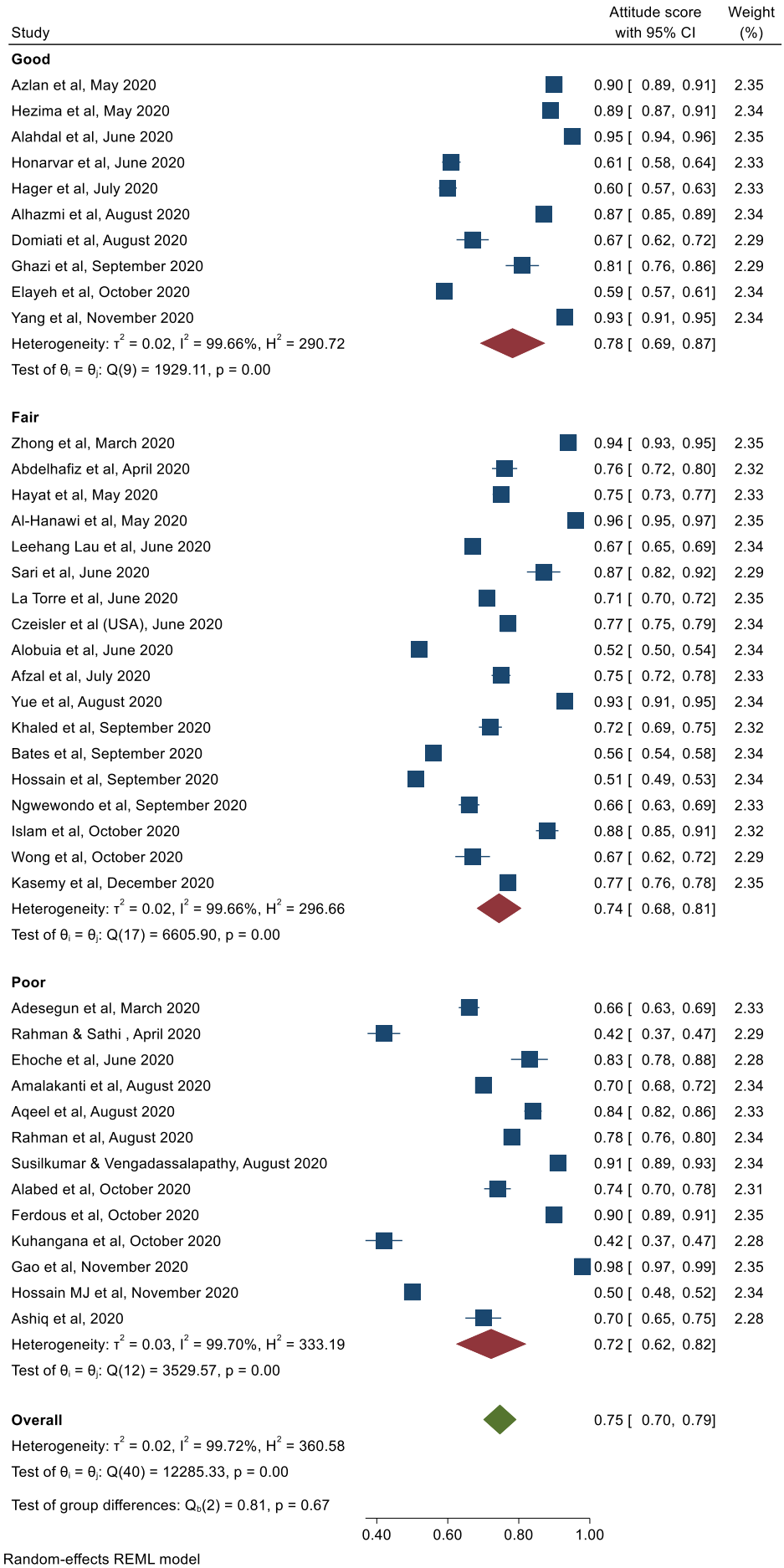


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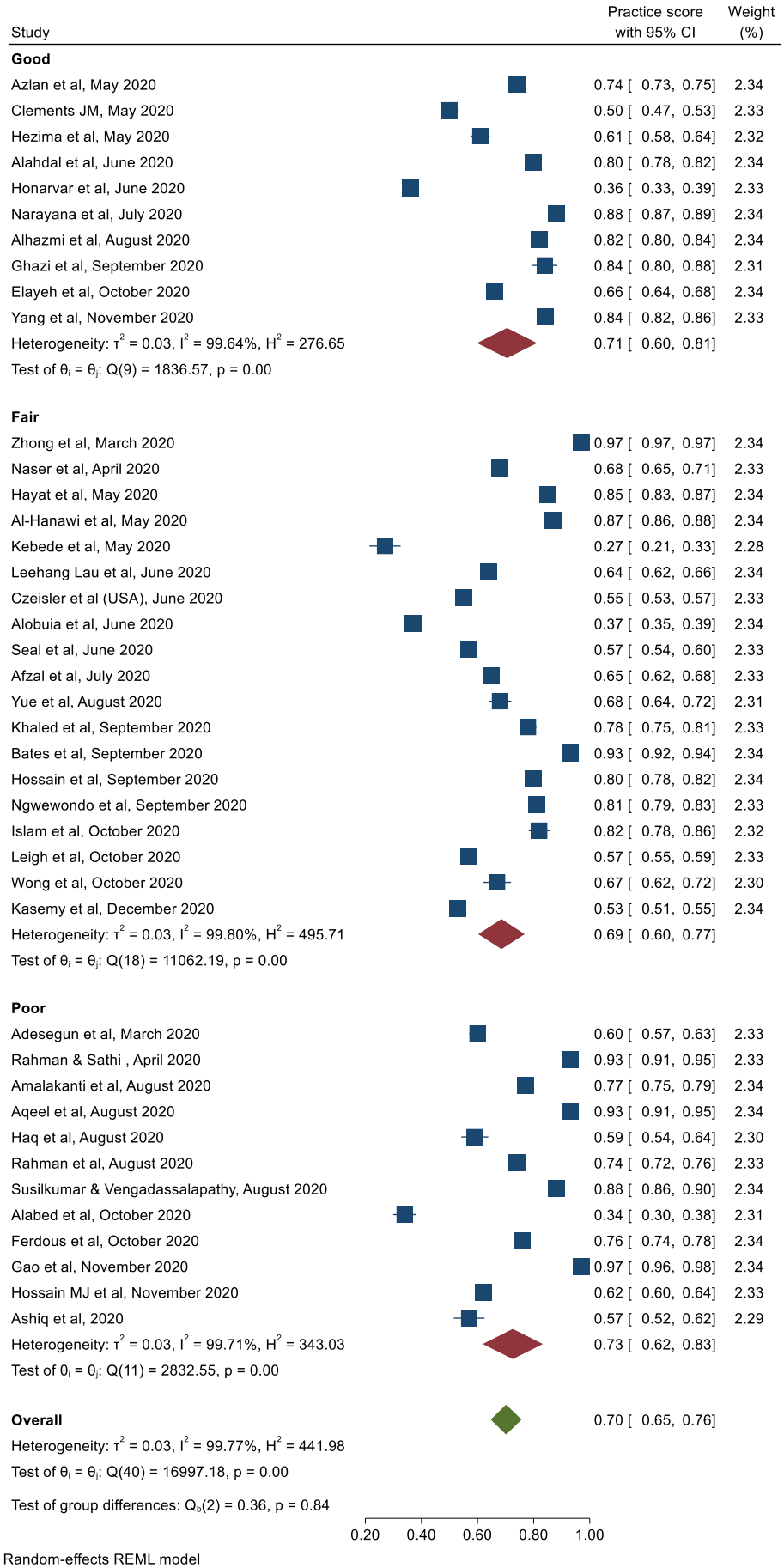
Figure 3: Analysis by quality of the studies



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Global epidemiology of COVID-19 knowledge, attitude and practice - a systematic review and meta-analysis

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Global epidemiology of COVID-19 knowledge, attitude and practice - a systematic review and meta-analysis

Objective
To assess the knowledge, attitude and practice (KAP) of the global general population regarding COVID-19.

Design
Systematic review and meta-analysis.

Methods
Medline, Embase, CINAHL and PsycINFO were used to identify articles published between 1 January and 30 June 2021 assessing KAP regarding COVID-19 in the global general population. The quality of eligible studies was assessed. Random effects model was used to obtain the pooled proportion of each component of KAP of COVID-19. Heterogeneity (I^2) was tested, and subgroup and correlation analyses were performed.

Results
Out of 3,099 records, 84 studies from 45 countries across all continents assessing 215,731 participants' COVID-19 KAP were included in this study. The estimated overall correct answers for knowledge, good attitude and good practice in this review were 75% (95% CI 72−77%), 74% (95% CI 71−77%) and 70% (95% CI 66−74%) respectively. Low-income countries, men, people aged below 30 years, and people with 12 years education or less had the lowest practice scores. Practice scores were below 60% in Africa and Europe/Oceania. Overall heterogeneity was high ($I^2 \geq 98\%$), and publication bias was present (Egger's regression test, $p<0.01$). A positive significant correlation between knowledge-practice ($r = 0.314$, $p = 0.006$), and attitude-practice ($r = 0.348$, $p = 0.004$) were observed.

Conclusions
This study's findings call for community-based awareness programmes to provide a simple, clear and understandable message to reinforce knowledge especially regarding efficacy of the preventive measures in low and lower-middle income countries, and in Africa and Europe/Oceania, which will translate into good practice. Targeted intervention for men, people with low education, unemployed people and people aged below 30 years should be recommended. As most of the included studies were online surveys, underprivileged and remote rural people may have been missed out. Additional studies are needed to cover heterogeneous populations.

PROSPERO registration number-CRD42020203476.

Strengths and limitations of this study
- This is the first systematic review and meta-analysis that summarises existing KAP pertaining to COVID-19 in the global general population.
- The study identified a large number of studies and participants, covering all continents of the world.
- Articles published in non-English were excluded, and substantial heterogeneity was present in the evaluated outcomes.
- As most of the included studies were cross-sectional online surveys, underprivileged and remote rural people who do not have internet access may have been missed out.

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Introduction

Emerging infectious diseases (defined as diseases that have 'newly appeared in a population or have existed previously but are rapidly increasing in incidence or geographic range'[1]) have always been a public health threat worldwide. Since late December 2019, the world has been facing Coronavirus Disease 2019 (COVID-19), which is caused by Severe Acute Respiratory Coronavirus 2 (SARS-CoV-2). On 30 January 2020, the World Health Organization (WHO) declared a state of public health emergency. Since then, this emerging and highly infectious disease has spread to 223 countries and territories, leading quickly to a global pandemic, with nearly 181.3 million cases and 3.9 million deaths reported as of 30 June 2021. At time of writing, COVID-19 is still challenging to contain for most countries around the world[2] and it mainly affects the respiratory system and can lead to acute hypoxaemic respiratory failure, and 0.9% to 14% of these patients require admission to intensive care units (ICU) for advanced respiratory support.[3 4] The consequent increase in critically ill COVID-19 patients with respiratory failure has overwhelmed intensive care units as well as the invasive mechanical ventilation capacity of healthcare systems in many countries across the globe.

Currently, vaccines are authorised and recommended to prevent COVID-19. Assuming two vaccine doses per person, according to a study published on 3 December 2020,[5] an estimated 15.6 billion doses of COVID-19 vaccines are needed for the 194 WHO member states to achieve a universal COVID-19 vaccination program. Furthermore, 10.3 billion doses are required for targeted and high-risk groups. According to 12 COVID-19 vaccine manufacturers across the world, about 10 billion doses will be available by the end of 2021. Vaccine production, however, is only the first step in a long delivery process that may face immense challenges such as logistics, cold chains and the actual administering of the vaccine.[5] Furthermore, according to WHO, about 50% of the vaccine will be wasted because of temperature control in the supply chains.[6] Therefore, distribution of vaccines, especially in rural and remote communities in many low- to lower-middle-income countries, requires strong international and national supply chains.[5] In addition, the literature shows that vaccine hesitancy may affect the benefits of vaccination campaigns,[7] and a recent global systematic review on COVID-19 vaccine hesitancy among general people showed that in many countries vaccine acceptance is below 60%.[8]

Thus, until the distribution of vaccine produces herd immunity, perceived-risk-induced behaviours and preventive measures work best to stop the further spread of the disease. This includes behaviours like avoidance of crowded places, washing hands regularly or using hand sanitisers, avoiding frequent touching of the eyes, nose and mouth, cleaning touched surfaces frequently, wearing a mask, social distancing and seeking medical advice upon noticing symptoms like sore throat, fever or shortness of breath.[9 10] People need to follow these measures voluntarily for the safety of themselves and other people in general. The level to which they follow them voluntarily will depend on their accurate knowledge about the pandemic and their attitude towards it.[11] However, since voluntary action is not enough, most governments across the world have been forced to come up with strict laws and measures that aim to stop the further spread of the disease. Still many countries across the world have not been successful in enforcing these laws. The biggest challenge with people's devotion to these control measures is determined by what they know about the disease, what their attitude is and what practices they undertake to follow their local measures.

Since the emergence of the virus, through the combined efforts of healthcare professionals and government bodies, several measures have been developed to raise awareness, improve knowledge and strengthen preventive practices to control the spread of infection.[12] However,

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lack of knowledge about mode of transmission and vulnerable populations, as well as lack of attention to preventive measures, may still be widespread; consequently, many countries are facing localised spikes or the localised return of a large number of cases and deaths.

The constant evolving nature of the COVID-19 pandemic necessitates the need for regular update of scientific literature and research techniques. While reviews have been published in the past on knowledge, attitude and practice (KAP), they are not comprehensive and needs timely update. One review article (that included articles published from December 2019 to September 2020) assessed predominantly the American sub-continent, and did not perform any definite or accurate quantitative analysis.[13] Another review article that included articles from March to July 2020, also did not perform any meta-analysis and included studies conducted on healthcare workers as well as on the general population.[14] Thus, the aim of the current study was to perform a systematic review and meta-analysis in order to provide the current global epidemiology of COVID-19 KAP in the general population. In comparison with the above published review studies, the present study was conducted over a more comprehensive time period (1 January 2020 to 30 June 2021) and performed a thorough meta-analysis with various in-depth sub-group analysis. To the best of our knowledge, the present study is the first global meta-analysis to have overall pooled prevalence of each component of KAP of COVID-19 among the global general population.

Methods

This systematic review and meta-analysis was conducted in accordance with 'Preferred Reporting Items for Systematic Reviews and Meta-analyses' (PRISMA).[15] It also complies with the Meta-Analysis and Systematic Reviews of Observational Studies (MOOSE) guidelines.[16] The study has been registered on PROSPERO (CRD42020203476). The review process is shown in the flow diagram in Figure 1 and the PRISMA and MOOSE checklist are included as Supplementary Table 1 and 2.

Eligibility criteria

Inclusion criteria: The review included studies that reported any form of quantitative assessment/measurement/evaluation of KAP regarding COVID-19 in the general population in any country or region across the world. There were no restrictions on the age, gender, ethnicity or health status of the participants, the duration for which the study was conducted or the geographic location of the study. Only published articles that had full text available (in English) were included.

Exclusion criteria: Studies were excluded if they were conducted only on some specific group of people, such as healthcare workers, medical students, pregnant women or people with co-morbidities. Short reports, case reports, briefs, letters, editorials and study duplicates were also excluded.

Search strategy, information sources and study selection

With the help of a senior medical librarian, two authors (BS and AS) independently searched the publicly available KAP in COVID-19-related published articles, using databases of MEDLINE, EMBASE, CINAHL and PsycINFO following the PRISMA and MOOSE checklist (Supplementary Table 1 and 2).[15] [16] This review also included pre-prints, which provide a dynamic update of research papers related to COVID-19. The main keywords for the search strategy included "knowledge", "attitude", "practice" and "COVID-19". Only English-language articles were considered, as a search restriction. EndNote was used to store, organise and manage all the citations.

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The same authors (BS and AS) then independently conducted the initial screening of the title and abstract for each article obtained from the literature search. When a title or abstract was not sufficient grounds to reject an article, the full text was obtained and assessed. After the independent review, they had a disagreement with the consideration of four papers (5%). Three papers were resolved by mutual consensus and one was resolved by consulting with the senior author (BB). Reference lists of the articles selected in this meta-analysis were also screened manually for any additional studies.

The preliminary search, which was carried out on 4 September 2020 and updated on 30 June 2021, covered published articles between 1 January 2020 (as the first novel case of COVID-19 was discovered on 31 December 2019) and 30 June 2021. The detailed search strategy is given in the Supplementary Table 3.

Study outcomes

This meta-analysis addressed three main outcomes of overall KAP pertaining to COVID-19 in the general population across the world.

Knowledge: Level of knowledge or awareness regarding symptoms, mode of transmission, people at risk, incubation and isolation periods, virus fatality, treatments and prevention of COVID-19 infection.

Attitude: Attitude towards controlling or managing of COVID-19.

Practice: Practices such as hand hygiene, wearing face masks, social distancing, avoidance of crowded places/social events and isolation/quarantine to prevent the spread of COVID-19.

The secondary outcomes included the estimated effect by age (under 30 years of age vs 30 or older), gender (male vs female), employment status (unemployed vs employed vs retired/student), income level of the country where the study was conducted (low-income vs lower-middle income vs upper-middle income vs high income), education level (up to 12 years vs over 12 years) continent**,** and study period (January-March 2020 vs April-June 2020 vs July-October 2020).

Data extraction

Using a data extraction template (Excel spreadsheet), two reviewers (BS and AS) independently extracted key information regarding identification of the study (authors, publication month, country where the studies were conducted), methodology (study setting, study design, study population, sample size, data collection tool, recruitment method, recruitment period), participants' demographics (gender, age, education, current employment status, income, area of residence) and main study findings (prevalence of correct knowledge/mean knowledge score of participants regarding symptoms, mode of transmission and prevention of COVID-19, prevalence of positive attitudes of participants towards controlling or managing of COVID-19 and prevalence of good practices such as social distancing, hand hygiene, wearing face masks and (where relevant) quarantine of participants to prevent COVID-19). The two data files were then cross-checked by a third reviewer (OB). Any disagreements were resolved by a consensus meeting under the supervision of a senior author (BB). Missing data, where required, was sought from the study authors.

Quality assessment and risk of bias in individual studies

The quality of the included studies was assessed by two independent reviewers (BS and AS) using the National Heart Lung and Blood Institute (NHLBI) quality assessment tool for observational cohort and cross-sectional studies.[17] The tool uses 14 criteria to assess internal validity and risk of bias. Each criterion was rated as 'yes', 'no', 'cannot determine', 'not

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applicable' or 'not reported' (Supplementary Table 4). An overall quality of the study was then rated as good, fair or poor. Any discrepancies in the rating between the two reviewers were resolved by a third author (OB).

Data Analysis

Quantitative data (e.g., percentage of KAP regarding COVID-19, and by age, gender, education level, current employment status of the participants) extracted from each study was analysed by the author AA and was then cross-checked by the senior author BB; any discrepancies were resolved. Statistical software package Stata-Version 16 (Statacorp, USA) was used for statistical analyses.

Means (standard deviations) or medians (25th and 75th percentiles), where appropriate, were extracted and then reported for numerical data. Percentage was extracted and then reported for each category of KAP, age group, gender, educational level and employment status of the study participants. The pooled proportion of COVID-19 KAP was calculated as a data synthesis of primary outcomes and was presented in forest plots. This analysis was performed using the random-effects model, as this method demonstrates better properties in the presence of heterogeneity (if any), accounting for both within-study and between-study variances.[18] Heterogeneity among the included studies was tested using the χ^2 test on Cochran's Q statistic, which was calculated by means of H and I^2 indices. The I^2 index represents the percentage of total heterogeneity across studies based on true between-study differences rather than on chance. I^2 values over 75% might represent substantial/considerable heterogeneity.[19] In order to identify the possible sources of substantial heterogeneity, subgroup analysis was conducted for each of the components of KAP by age group, gender, education level, employment status, income level of the country, continent and study period.[20] The income level of each country was classified as low-income, lower-middle income, upper-middle income or high income, according to the World Bank.[21] A comparison of China vs other countries was done because China was the epicentre of COVID-19, and was the first country to introduce the preventive measures and strict laws. Hence, we wanted to analyse if this difference in timeline would have an effect on its people's knowledge, attitude and practice in comparison with other countries. Egger's regression test was used to examine for publication bias, and the symmetry of the funnel plots was evaluated.[22] Sensitivity analyses were performed using quality score categories of the included studies. Furthermore, correlation analysis among the components of KAP was also performed to evaluate the strength and direction of correlation (denoted by r) between knowledge and attitude, attitude and practice, and knowledge and practice.

Patient and public involvement

There was no patient or public involvement in this systematic review of published literature.

Results

A total of 3,099 articles, published between 1 January 2020 and 30 June 2021, were retrieved from the four databases and manual searches. After removing duplicate records and screening by titles and abstracts, 126 articles were included for full-text reading. Of these, 42 articles were excluded by the exclusion criteria. Finally, 84 studies from 45 countries or territories, reporting 215,731 participants' KAP towards COVID-19, were included in the qualitative and quantitative analysis.[11 23-105] Of them, 66 were cross-sectional online surveys, 12 were community-based, five were both community-based and online surveys and one was a hospital-based cross-sectional study. Based on the National Heart Lung and Blood Institute (NHLBI) quality assessment tool, 17 studies were of good quality, 44 were of fair quality and the remaining 23 were of poor quality. Supplementary Table 5 summarises the study features and

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the characteristics of the participants. The age of the study participants ranged between 16 and 75 years, and 41.7% were male. In the context of participants' education, 23.8% had up to secondary-level education, and 74.3% had university-level education. About 48.7% of participants were employed, 15.2% were unemployed, 14.3% were students and 3.5% were retired. In terms of country income level, 18% of the studies (n = 32,617) were from high-income countries, 30% (n = 63,545) were from upper-middle-income countries, 32% (n = 33,360) were from lower-middle-income countries and 20% (n = 13,293) were from low-income countries.

Figure 2 represents the pooled proportion of people who had answered correctly in their knowledge questions, had positive attitude and good practice techniques. The findings demonstrated that 75% of the present study population answered the knowledge question correctly (95% CI 72–77%, $p < 0.001$, $I^2 = 99.55\%$), 74% had positive attitude towards COVID-19 (95% CI 71–77%, $p < 0.001$, $I^2 = 99.58\%$), and 70% followed good COVID-19 practice techniques (95% CI 66–74%, $p < 0.001$, $I^2 = 99.69\%$). High heterogeneity was observed ($I^2 > 99\%$), with the presence of publication bias (Egger's regression test, $p < 0.01$). Correlation analysis among the components of KAP was also performed and is presented in Supplementary Figure 1. Very weak and insignificant correlation of knowledge with attitude was observed. However, practice had a significant positive correlation with both knowledge ($r = 0.314$, $p = 0.006$) and attitude ($r = 0.348$, $p = 0.004$). Furthermore, practice measures pertaining to COVID-19 did not correlate with time since the emergence of this pandemic. Prevalence of each component of KAP by countries has been presented in Figure 3.

Supplementary Table 6 presents the subgroup analysis results by age group, gender, education level, employment status, country income level, study period, and continents. Respondents aged 30 years and above had relatively higher knowledge (80%, 95% CI 75−85% vs 78%, 95% CI 74−83%), attitude (73%, 95% CI 65−81% vs 68%, 95% CI 59−78%) and practice (82%, 95% CI 74−90% vs 80%, 95% CI 72−88%) compared to those aged below 30. However, the difference was not significant as the respective confidence intervals overlapped. There were no significant differences between male and female participants regarding knowledge, attitude and practice, however practice scores were slightly lower among male participants (75%, 95% CI 69−80%) than females (77%, 95% CI 72−82%). Participants with 12 years of education or less had non-significant lower knowledge and practice than, but similar attitudes to, those with above 12 years of education. There was no significant difference between employed and unemployed participants when compared by each component of KAP. Knowledge score ranged between 72% (95% CI 63−80%) for low income countries and 79% (95% CI 75−83%) for upper-middle income countries. There was no progressive increase or decrease in attitude and practice scores by income level. However, low income countries had the lowest attitude score (69%, 95% CI 62−76%) and that was the highest for upper-middle income countries (78%, 95% CI 72−84%). In the context of practicing scores, low-income countries had the lowest (55%, 95% CI 46−64%), and the highest came from the upper-middle income countries (77%, 95% CI 70−84%) and the difference was statistically significant. Analysis by continent showed non-significant differences. However, Americas had the highest knowledge score (79%, 95% CI 74−83%) and that was the lowest in Europe/Oceania (67%, 95% CI 54−80%). Attitude scores were highest in Middle-East (79%, 95% CI 72−85%) and lowest in Americas (66%, 95% CI 56−77%). Asia had the non-significant highest practice score (76%, 95% CI 71−81%) as compared to other countries. Further subgroup analysis was done by time of data collection (study period) into three months interval (January-March 2020, April-June 2020, July-October, 2020). There was no significant increase or decrease in knowledge and practice over time. However, positive attitude towards COVID-19 had been decreased significantly over the time

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(January-March 2020: 80%, 95% CI 74−86%; April-June 2020: 73%, 95% CI 68−77%; July-October 2020: 65%, 95% CI 57−73%). Heterogeneity was high ($I^2 > 98\%$) across all subgroup analyses (except for Europe/Oceania for attitude and practice where the heterogeneity was 0.29% and 0.28% respectively), and publication bias was present in most of the studies.

The analyses comparing KAP between studies from China versus other counties are illustrated in Supplementary Figure 2. The KAP scores across five studies from China (n = 10,482) reported an overall higher prevalence of each component of KAP of 79% (95% CI 67–92%), 92% (95% CI 86–97%) and 85% (95% CI 74–96%) as compared to studies from other countries of 74% (95% CI 71–77%), 72% (95% CI 69–76%) and 69% (95% CI 65–73%) respectively, which showed statistical significance for attitude and practice. Sensitivity analyses were performed by quality of the studies, as presented in Supplementary Figure 3; however, high heterogeneity and presence of publication bias still existed.

Further analysis was also done on some important individual questions regarding knowledge, attitude and practices towards COVID-19, as illustrated in Supplementary Table 7. Of all participants, 85% and 81% had correct knowledge on 'main symptoms' (reported by 61 studies) and 'population at risk' (reported by 46 studies) of COVID-19 respectively, while 83% (reported by 58 studies) and 75% (reported by 39 studies) of respondents were aware of 'route of transmission' and spread of infection by asymptomatic patients, respectively. Knowledge regarding avoidance of crowded places as a preventive measure was reported by 27 studies, with overall correct knowledge of 95%. Correct knowledge about 'isolation and treatment of patients' (reported by 32 studies) and 'quarantine or incubation period' (reported by 44 studies) of COVID-19 was observed among 91% and 87% of the study participants respectively. However, only 70% of the participants, reported in 35 studies, had correct knowledge about 'wearing a face mask' as a preventive measure.

Positive attitude towards 'successful control of COVID-19' (reported by 22 studies) and 'social distance' (reported by 11 studies) was reported by 71% and 88% of the study participants respectively.

Practicing hand hygiene through 'proper hand washing or using hand sanitiser' was reported by 80% of the participants in 49 studies; 70% of the study participants (reported by 32 studies) maintained social distance, and 75% (reported by 45 studies) avoided crowded places or social events to prevent the spread of infection. Unexpectedly, only 65% of the participants (reported by 56 studies) were practicing the wearing of face masks while going out. High heterogeneity was observed in all these subgroup analyses, and presence of publication bias was observed in most.

Discussion

This is a large international systematic review and meta-analysis that has examined the knowledge, attitude and practice of the general population on the novel disease COVID-19. Data was compiled from 215,731 participants from 84 studies of 45 countries, covering all continents. The study demonstrated a pooled score of ≥70% in each of the three components of KAP. The overall practice of preventive measures was low in the low-income countries, in Africa and Europe/Oceania, and among men, employed people and people aged 30 and above. There was a positive correlation between the components of KAP.

The estimated overall correct answers for knowledge, good attitude and good practice techniques in this review were 75%, 74% and 70% respectively. A study from H1N1 influenza

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A reported a lower knowledge (60%) and similar attitude (74.5%), but higher practice (77.3%) scores, compared to current study.[106] Studies from Severe Acute Respiratory Syndrome (SARS-CoV) in 2003 reported a knowledge score of 52.4–67.4% on symptoms among the general population compared to 85% in our study.[107] Furthermore, a study from Saudi Arabia on Middle East Respiratory Syndrome (MERS) reported study participants' fair knowledge on symptoms (79.8%) and transmissions (90.6%), and their practice on 'washing hands frequently' (94%) and 'using face masks' (74.9%),[108] which were higher than those in the current study. This could be due to high risk perception and their longer experience with MERS (as the study was conducted two years after the first appearance of MERS-CoV).

Subgroup analysis showed that overall knowledge score was low in the low and high income countries, in Europe/Oceania, among participants with 12 years or less education and unemployed people, and attitude score was low in Africa, Americas and low income countries. Low practice score was observed in Africa, Americas, Europe/Oceania, the low income countries, and among males, employed people, people aged below 30 years, and participants with 12 years of education or less. A meta-analysis conducted among general population to see 'the association between gender and protective behaviours in response to respiratory epidemics and pandemics' revealed that female are about 50% more likely than male to adopt or practice preventive behaviours (e.g., hand washing, face mask use, avoidance of public transport).[109] Education attainment may also have a direct effect on knowledge and consequently on practices as people with less education are less likely to adopt preventive measures. Population density, living conditions, livelihood and affordability could be the reasons behind low practices in the low income countries. Moreover, due to existing health inequalities in low-income and lower-middle-income countries and in socially disadvantaged groups, pandemics generally have their greatest effects on these populations.[5] During the 1918 "Spanish" influenza pandemic and in the 2009 H1N1 influenza pandemic, racial minority/minority groups were most affected.[110 111] However, literature supports that during COVID-19 pandemic, more cases and deaths have been reported from high-income and upper-middle-income countries.[5] These differences may be related to timing of transmission, countries' response patterns, availability and accuracy of tests, socio-demographic condition, countries' environmental factors and connectivity and trade patterns.

Analysis by study period showed no progressive increase or decrease in knowledge over time, however the positive attitude towards COVID-19 had been decreased significantly over time (January 2020 to October 2020). A major reason for this may be attributed to pandemic fatigue wherein the prolonging timeline of the pandemic makes it harder to follow the COVID-19 prevention guidelines. Besides, a decrease in perceived susceptibility and perceived severity may further deteriorate the positive attitude towards the pandemic.[112] The practice of COVID-19 prevention measures also did not increase since the emergence of this pandemic in early 2020, which may be partly reflected from the multiple spikes of COVID-19 spread in many countries across the world. This may be due to lack of understanding of the severity of the disease or true risk, low perceived risk of acquiring infection and perceived efficacy of the preventive measures.

Further subgroup analysis showed that overall knowledge, attitude and practice scores were higher (79%, 92% and 85% respectively) in the studies from China, the early COVID-19 epicentre, compared to studies from other countries 74%, 72% and 69%, respectively). This could be due to highly educated study participants in the study from Wuhan (over 80% had associate degrees or higher), overwhelming news by the media and public health authorities,

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their experience with previous severe acute respiratory syndrome outbreak, and strict actions taken by the local government to control the infection immediately after the outbreak.

In this current study, the overall score of knowledge regarding individual questions about main symptoms, population at risk, route of transmission of COVID-19, isolation and treatment of infected people, and quarantine time was higher than 80%; however, overall, 75% and 70% of respondents had knowledge of asymptomatic patients and the use of face masks as preventive measures, respectively. There were significant differences between knowledge and actual practices, and attitude and practices – specifically, avoidance of crowded places and maintaining social distance. Though 95% of participants had the knowledge of avoidance of crowded places to prevent the infection, only 75% of them were practicing it. On the other hand, 88% of participants had positive attitude to maintain social distance, only 70% of them were practicing it. This difference could be not only due to lack of understanding of the preventive measures, the rationales for their use and their impact on health, but also due to population density and livelihood. Studies from H1N1 influenza A reported correct knowledge on symptoms (68.1–76%), route of transmission (56.3–75.6%), use of face mask (36.6–70.4%), and practice of frequent hand washing (54.6–56.9%) and avoidance of crowded places (42.9–52.6%), which were lower than those in this current study.[113-115] This could be due to low education level (nearly 50% of study participants had below primary/middle school education), lack of perceived risk of spread of the disease and lack of perceived efficacy of various preventive measures among the H1N1 study participants.

A correlation analysis conducted in this study showed a positive relationship between components of KAP. Literature supports correlation between knowledge, attitude and practice in relation to health issues, which supports this study findings.[116]

A wide heterogeneity was observed in the reported prevalence of each component of KAP, ranging between 35–95%, 42–96% and 27–97%, respectively. This was partly explained by the socio-demographic and cultural differences and political influence, and partly by the differences in the questionnaire contents, measurement and scoring systems. The majority of the studies were online surveys, and literature supports that gender, age and education influence online survey behaviour.[27] Internet and social media are today's important online health information resources, hence their users may have gained more information than underprivileged population. Differences in the time and actions taken by the government, trust in the governing institutions and experience in management of previous outbreaks may have influenced the attitude of the participants.[27 34 45 70] Mixed message about the use of face masks in public places, scarcity and affordability of face masks, cultural norms to shake hands and going to social, family or religious gatherings, continuity of water supply and hand washing facilities and living conditions may have contributed to adopt the preventive practices against COVID-19 infection.[33 46 52 117]

This systematic review has several strengths and limitations. First, because of the challenges in conducting research during the pandemic, most of the studies were online surveys and covered major cities of the respected countries. Therefore, vulnerable population particularly income-poor people and urban slum dwellers, and people living in remote rural areas who did not have internet access may have missed out. Second, heterogeneity was high in the evaluated outcomes. Although a range of subgroup analyses have been performed, the sources of heterogeneity could not be identified. Third, articles published in other than English were excluded, which may have resulted in the exclusion of some studies. Despite these limitations, to our knowledge, this is the first systematic review and meta-analysis on KAP of COVID-19

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among global general population. Furthermore, the study identified a large number of studies and participants, covering all continents of the world, and provided meaningful data on the KAP of COVID-19 in the general population.

Conclusion

The overall KAP pertaining to COVID-19 in the general population in this study was 75%, 74% and 70% respectively. Low practice score was observed in Africa and Europe/Oceania, in low income countries, and among men, employed, aged below 30 years, and participants with 12 years of education or less. This data suggest targeted group intervention to reinforce knowledge giving attention to efficacy of each of the preventive measures, which will translate into good practice. Additional studies covering heterogeneous population are also needed.

Figure legend

Figure 1: PRISMA 2009 Flow Diagram
Figure 2: Pooled proportion of knowledge, attitude and practice of COVID-19
Figure 3: Geographical representation of knowledge, attitude and practice during the COVID-19 pandemic

Contributors

BB, AA and BS conceived the study and together with AS and OB developed the protocol. BS and AS did the literature search and selected the studies. BS and AS extracted the data, and OB verified the data. AA and BB synthesised the data. BS drafted the initial report with input from AS and BB. BS, AS, OB, AA and BB critically revised successive drafts of the paper. All authors approved the final version of the report.

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Declaration of interests

We declare no competing interests.

Patient consent for publication

Not required.

Ethics approval

Ethics approval was not required because the analyses in this review are from data that already publicly available in published articles.

Data availability statement

All data generated or analysed during this study are included in this article and its supplementary information files.

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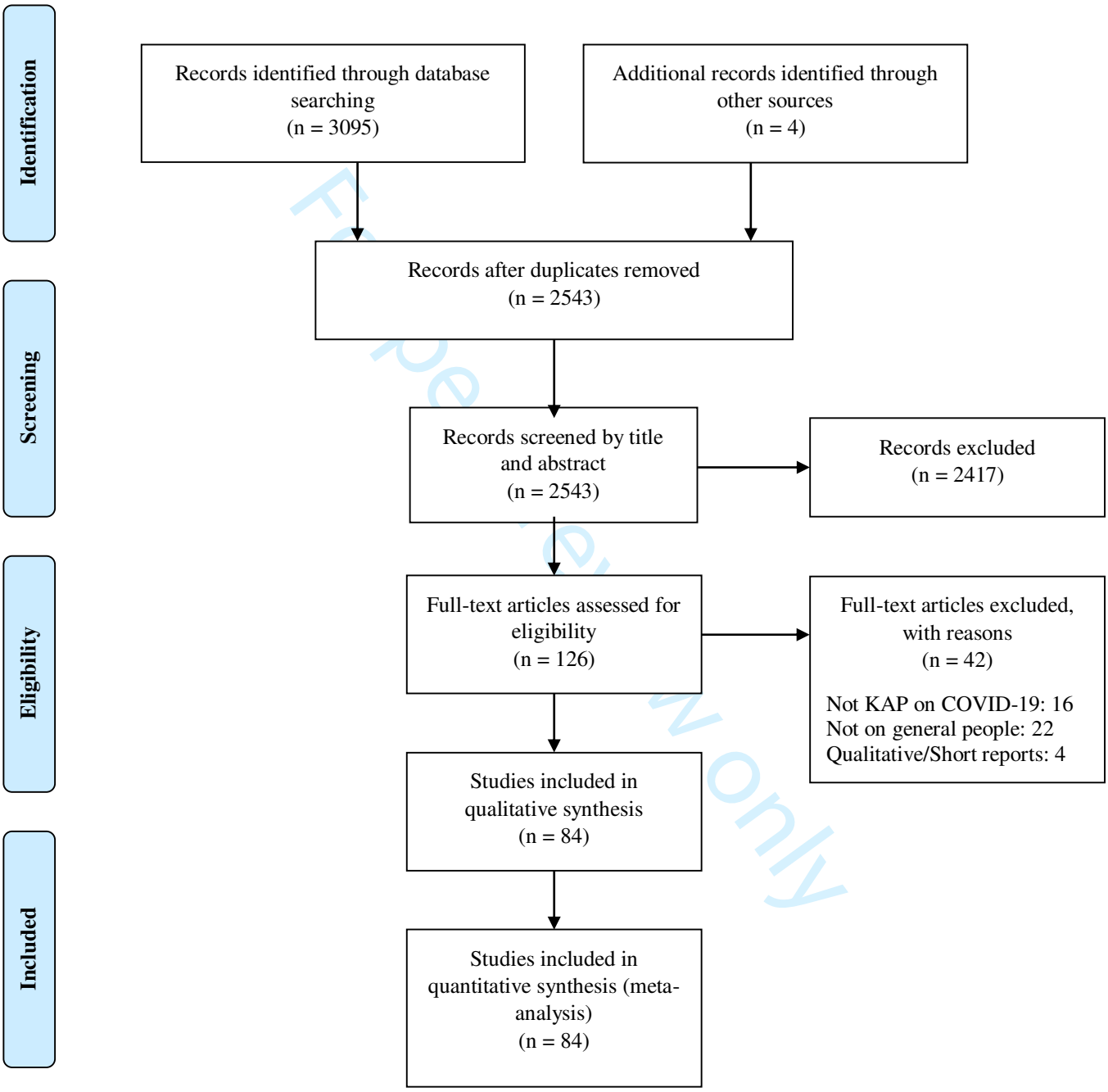
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Figure 1: PRISMA 2009 Flow Diagram



Note: KAP: Knowledge, attitude and practice

Figure 2: Pooled proportion of knowledge, attitude and practice of COVID-19

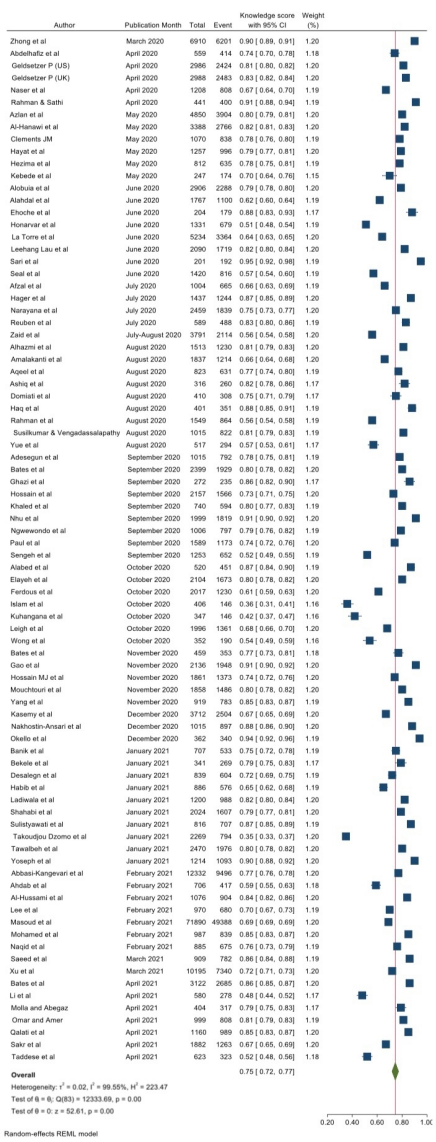


Figure 2a: Pooled proportion of knowledge of COVID-19

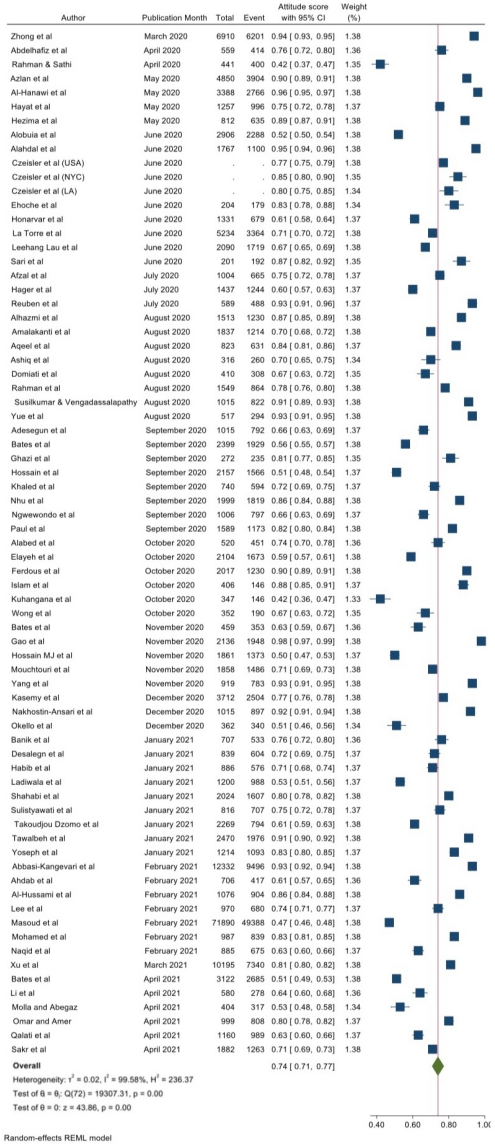


Figure 2b: Pooled proportion of attitude of COVID-19

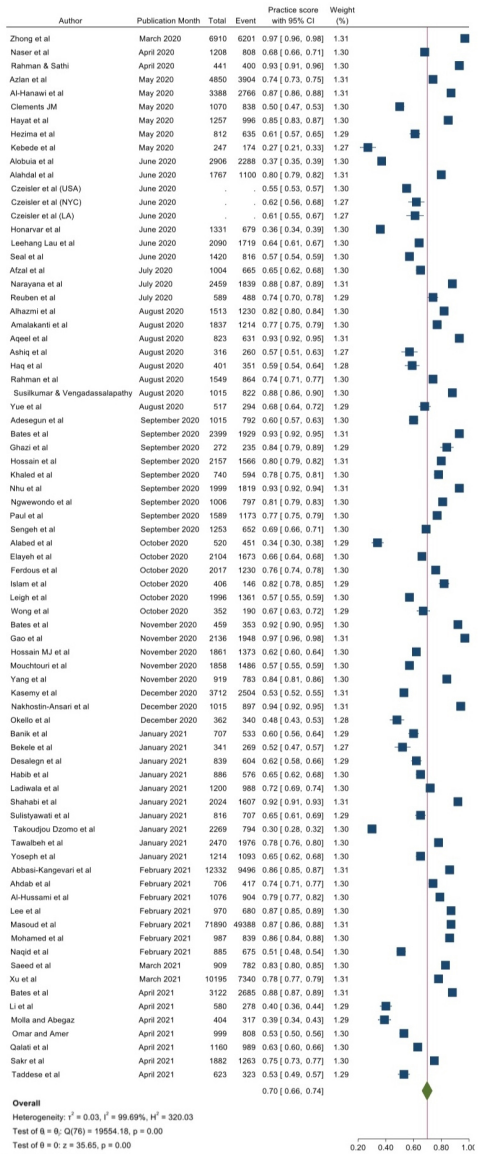
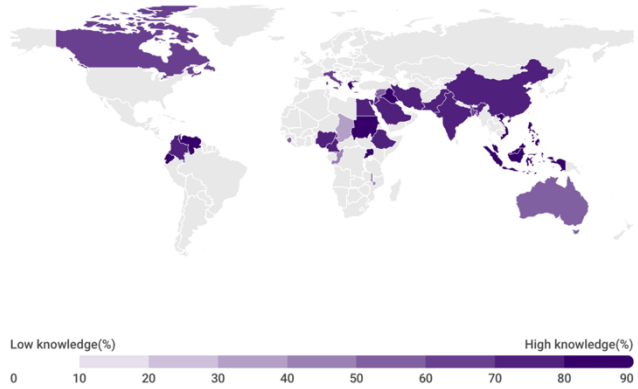


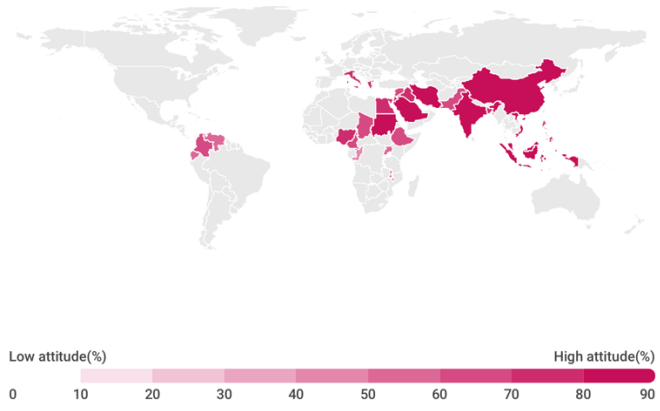
Figure 2c: Pooled proportion of practice of COVID-19

Figure 3: Geographical Representation of Knowledge, Attitude and Practice during the COVID-19 pandemic

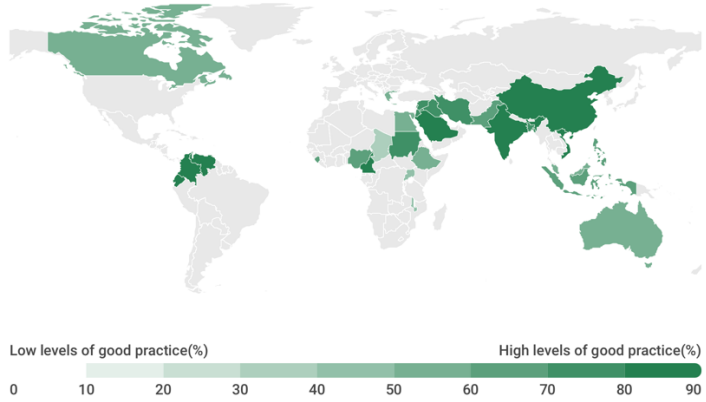
3A: Geographical representation of knowledge levels during the COVID-19 pandemic.



3B: Geographical representation of attitude levels during the COVID-19 pandemic



3C: Geographical representation of practice levels during the COVID-19 pandemic



Supplementary material

Global epidemiology of COVID-19 knowledge, attitude and practice - a systematic review and meta-analysis

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Appendix

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Table 1: PRISMA Checklist[15]

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Page 4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Page 4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Page 4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Page 4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Page 4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary Table 3
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Page 4-5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Page 5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Page 5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Page 5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Page 6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Page 6
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Page 6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Page 6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Page 6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Supplementary Table 5
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Page 6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Page 7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Page 7-8
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Page 7-8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Page 7-8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Page 8
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Page 10
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Page 11
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Page 11

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Table 2: MOOSE Checklist[16]

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	3
2	Hypothesis statement	-
3	Description of study outcome(s)	5
4	Type of exposure or intervention used	-
5	Type of study designs used	4
6	Study population	4-5
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	Title page
8	Search strategy, including time period included in the synthesis and key words	4
9	Effort to include all available studies, including contact with authors	4-5
10	Databases and registries searched	4
11	Search software used, name and version, including special features used (eg, explosion)	-
12	Use of hand searching (eg, reference lists of obtained articles)	5
13	List of citations located and those excluded, including justification	6, 13-17, Fig 1
14	Method of addressing articles published in languages other than English	-
15	Method of handling abstracts and unpublished studies	-
16	Description of any contact with authors	5
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	-
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	-
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	5
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	-
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	5
22	Assessment of heterogeneity	6
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	6-7
24	Provision of appropriate tables and graphics	Tables 1-4, Figs 1
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	Figs 2, 3
26	Table giving descriptive information for each study included	Supplementary Table 5
27	Results of sensitivity testing (eg, subgroup analysis)	Suppl. Table 6
28	Indication of statistical uncertainty of findings	7-8
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	10
30	Justification for exclusion (eg, exclusion of non-English language citations)	10
31	Assessment of quality of included studies	-
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	9-10
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	-
34	Guidelines for future research	12
35	Disclosure of funding source	12

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Table 3: Search strategy in MEDLINE

	Search terms
#1	(wuhan and (coronavirus or corona virus)).tw,kf,hw.
#2	(coronavirus* and ("19" or "2019")).tw,kf,hw.
#3	(2019 nCov or nCov 2019 or ncov19 or ncov 19 or novel coronavirus* or novel corona virus* or Severe Acute Respiratory Syndrome Coronavirus 2 or coronavirus disease 2019 or corona-virus disease 2019 or new coronavirus* or new corona-virus* or SARS-Coronavirus-2 or SARS-Coronavirus2 or SARS-Corona-Virus-2 or SARS-corona-virus2 or SARS-like coronavirus*).tw,kf,hw.
#4	(2019-novel CoV or SARS-COV-2 or SARS-COV2 or sarscov2 or sarscov-2 or coronavirus-19 or covid19 or covid-19).tw,kf,hw.
#5	((novel or new or nouveau or pandemic*) adj2 (CoV or covid or ncov or coronavirus or corona-virus)).tw,kf,hw.
#6	#1 OR #2 OR #3 OR #4 OR #5
#7	Knowledge/
#8	Attitude/
#9	Practice/
#10	Health Knowledge, Attitudes, Practice/
#11	#7 OR #8 OR #9 OR #10
#12	#6 AND #11
#13	limit 12 to english language

Table 4: Tool used for quality assessment[17]

	Criteria	Yes	No	Others (CD, CR, NA)*
1	Was the research question or objective in this paper clearly stated?			
2	Was the study population clearly specified and defined?			
3	Was the participation rate of eligible persons at least 50%?			
4	Were all the subjects selected or recruited from the same or similar populations (including the same time period)? Were inclusion and exclusion criteria for being in the study pre-specified and applied uniformly to all participants?			
5	Was a sample size justification, power description, or variance and effect estimates provided?			
6	For the analyses in this paper, were the exposure(s) of interest measured prior to the outcome(s) being measured?			
7	Was the time frame sufficient so that one could reasonably expect to see an association between exposure and outcome if it existed?			
8	For exposures that can vary in amount or level, did the study examine different levels of the exposure as related to the outcome (e.g., categories of exposure or exposure measured as continuous variable)?			
9	Were the exposure measures (independent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?			
10	Was the exposure(s) assessed more than once over time?			
11	Were the outcome measures (dependent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?			
12	Were the outcome assessors blinded to the exposure status of participants?			
13	Was loss to follow-up after baseline 20% or less?			
14	Were key potential confounding variables measured and adjusted statistically for their impact on the relationship between exposure(s) and outcome(s)?			
	Quality rating (good, fair, or poor) (see guidance) Rater no. 1 initials: Rater no. 2 initials: Additional comments (if poor, please state why):			

∗CD: cannot determine; NA: not applicable; NR: not reported.

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Table 5: Studies included in the systematic review and meta-analysis

	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
Asia	Azlan et al, May 2020[33]	Malaysia	Cross-sectional online survey	Malaysian above 18 years	4850	34.0	42.1	Good	3904	80.0	4346	90.4	3564	74.1
	Afzal et al, July 2020[25]	Pakistan	Cross-sectional online survey	Pakistani nationals aged 16 years or more	1004	NR*	37.0	Fair	665	66.2	756	75.3	654	65.2
	Hayat et al, May 2020[45]	Pakistan	Cross-sectional online survey	Pakistani residents age >15yrs	1257	16–30+	44.3	Fair	996	79.2	949	75.5	1073	85.3
	Leehang Lau et al, June 2020[56]	Philippines	Community-based Cross-sectional survey	participants of ICM's poverty alleviation program	2224	41.3	7.3	Fair	1719	82.2	1404	67.2	1328	63.5
	Rahman & Sathi, April 2020[61]	Bangladesh	Cross-sectional online survey	Bangladeshi internet users ≥18 years	441	18–30+	68.7	Poor	400	90.7	184	41.7	408	92.6
	Sari et al, June 2020[63]	Indonesia	Cross-sectional online survey	aged 18–60 years	201	35.6	46.3	Fair	192	95.5	174	86.6	NR	NR
	Yue et al, August 2020[68]	Henan, China	Cross-sectional online survey	People aged 15 years or older	517	15–60	46.2	Fair	294	56.9	482	93.3	350	67.7
	Zhong et al, March 2020[70]	China	Cross-sectional online survey	Chinese nationality aged 16 years or more	6910	33.0	34.3	Fair	6201	89.7	6492	93.9	6717	97.2
	Alabed et al, October 2020[26]	Malaysia	Cross-sectional online survey	People living in Malaysia aged 18 years or above	520	36.9	38.5	Poor	451	86.7	386	74.3	178	34.2
	Amalakanti et al, August 2020[30]	Andhra Pradesh, India	Cross-sectional online survey	South Indian aged above 15 years	1837	16–50+	56.5	Poor	1214	66.1	1286	70.0	1414	77.0
	Aqeel et al, August 2020[31]	Delhi, India	Cross-sectional online survey	Residents of Delhi	823	38.2	56.9	Poor	631	76.6	695	84.4	769	93.4
	Ashiq et al, August 2020[32]	Lahore, Pakistan	Cross-sectional online survey	People residing in Lahore	316	16–40+	46.5	Poor	260	82.3	221	69.9	181	57.2
	Ferdous et al, October 2020[40]	Bangladesh	Cross-sectional online survey	Bangladeshi residents aged 12-64 years	2017	24.4	59.8	Poor	1230	60.9	1818	90.1	1534	76.1

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Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
								Event, n	%	Event, n	%	Event, n	%
Gao et al, November 2020[71]	China	Cross-sectional online survey	Chinese residents 18 years and above	2136	33.1	21.9	Poor	1948	91.2	2093	98.0	2068	96.8
Haq et al, August 2020[44]	Punjab, Pakistan	Cross-sectional online survey	People living in Pakistan >15 years	401	NR	52.3	Poor	351	87.6	NR	NR	236	58.7
Hossain et al, September 2020[48]	Bangladesh	Cross-sectional online survey	Bangladeshi residents 13-90 years	2157	33.5	54.1	Fair	1566	72.6	1109	51.4	1716	79.5
Hossain MJ et al, November 2020[49]	Bangladesh	Cross-sectional community-based and online study	Bangladeshi residents	1861	15−50+	64.5	Poor	1373	73.8	936	50.3	1154	62.0
Islam et al, October 2020[50]	Bangladesh	Cross-sectional community-based study	Aged ≥18 years slum dwellers	406	44.9	53.2	Fair	146	35.9	357	87.9	332	81.7
Narayana et al, July 2020[58]	India	Cross-sectional online survey	Indian residents aged above 15 years	2459	24.5	57.9	Good	1839	74.7	NR	NR	2167	88.1
Rahman et al, August 2020[61]	Dhaka, Rangpur and Bogura, Bangladesh	Cross-sectional online and offline study	Adult population living in Bangladesh	1549	18−61	57.9	Poor	864	55.8	1207	77.9	1140	73.5
Susilkumar & Vengadassalapathy, August 2020[65]	India	Cross-sectional online survey	Participants with access to a primary internet connection and who can read and write the English language	1015	20−60	49.3	Poor	822	81.0	926	91.1	892	87.8
Wong et al, October 2020[66]	Hong Kong	Cross-sectional study	Age of 18 years or older South Asian in Hong Kong	352	38.9	40.3	Fair	190	53.8	238	67.4	234	66.5

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	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
	Yang et al, November 2020[67]	China	Cross-sectional online survey	Residents who were aged 18 years and above, understood Chinese	919	18−46+	21.7	Good	783	85.2	854	92.9	776	84.4
	Nhu et al, September 2020[103]	Vietnam	Cross-sectional online survey	Vietnamese population	1999	18−59	21.7	Fair	1819	90.6	1719	86.0	1845	93.4
	Paul et al, September 2020[92]	Bangladesh	Cross-sectional online survey	Bangladeshi citizens aged 18 and older	1589	18−45+	60.5	Fair	1173	73.8	1303	81.6	1224	77.3
	Banik et al, January 2021[75]	Bangladesh	Cross-sectional online survey	Bangladeshi youth aged 18-35 years	707	25.03±4.26	57.1	Good	533	75.4	541	76.4	427	60.4
	Ladiwala et al, January 2021[81]	Pakistan	Cross-sectional online survey	aged 15 years and above who were permanent residents of Pakistan	1200	15−60+	38.3	Fair	988	82.3	631	52.5	859	71.6
	Sulistyawati et al, January 2021[99]	Indonesia	Cross-sectional online survey	Residence of Indonesia aged 18 years or older	816	18−50+	27.0	Poor	707	86.7	612	75.0	533	65.3
	Lee et al, February 2021[82]	South Korea	Cross-sectional online survey	aged 18 years or older, resident in South Korea and Korean speaker	970	47.44±14.78	48.6	Fair	680	70.2	718	74.0	844	87.0
	Xu et al, March 2021[104]	China	Cross-sectional online survey	Residents of 32 provinces of China	10195	30.2 ± 8.5	44.6	Fair	7340	72.4	8258	80.5	7952	77.5
	Qalati et al, April 2021[93]	China, India, Pakistan	Cross-sectional online survey	General public of China, India and Pakistan	1160	NR	71.7	Fair	989	85.3	731	63.1	731	63.4
Africa	Abdelhafiz et al, April 2020[23]	Egypt	Cross-sectional community-based and online survey	Egyptian adults	559	18−60+	37.7	Fair	414	74.1	424	75.8	NR	NR
	Ehoche et al, June 2020[38]	Nigeria	Cross-sectional online survey	North Central Nigeria	204	15−60	58.8	Poor	179	87.6	169	82.8	NR	NR

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								Event, n	%	Event, n	%	Event, n	%
Hezima et al, May 2020[46]	Khartoum, Sudan	Cross-sectional community-based and online survey	Sudanese residents	812	18−55+	54.2	Good	635	78.2	725	89.3	494	60.8
Hager et al, July 2020[43]	Egypt and Nigeria	Cross-sectional online survey	adults >17 years of all educational levels	1437	18−59+	52.5	Good	1244	86.6	863	60.0	NR	NR
Kebede et al, May 2020[52]	Jimma town, South-west Ethiopia,	Hospital-based Cross-sectional study	Visitors of medical centre	247	30.5	76.5	Fair	174	70.4	NR	NR	66	26.9
Adesegun et al, September 2020[24]	Nigeria	Cross-sectional online survey	Literate nigerian who understand english	1015	26.6	45.9	Poor	792	78.0	670	66.0	613	60.4
Kasemy et al, December 2020[51]	Egypt	Cross-sectional online survey	Egyptians	3712	23.3	47.8	Fair	2504	67.4	2856	76.9	1950	52.5
Kuhangana et al, October 2020[54]	Democratic Republic of the Congo	Cross-sectional study	adult people (>18 years) frequenting ten public markets	347	37.4	17.0	Poor	146	42.2	147	42.5	NR	NR
Ngwewondo et al, September 2020[60]	Cameroon	Cross-sectional online survey	Cameroonian residents, aged 18 years or more	1006	33.0	46.9	Fair	797	79.2	663	65.8	812	80.7
Reuben et al, July 2021[94]	North-Central Nigeria	Cross-sectional online survey	Residents of north-central Nigeria who understood the English language and were 18 years old and above	589	18−59	59.6	Fair	488	82.8	545	92.5	435	73.9
Sengeh et al, September 2020[97]	Sierra Leone	Community-based cross-sectional study	Resident aged 18 years or older	1253	18−60+	52.0	Fair	652	51.6	NR	NR	865	68.7
Okello et al, December 2020[91]	Uganda	Cross-sectional online survey	Ugandans aged ≥ 18 years	362	33.5 ± 10.4	58.6	Poor	340	93.9	186	51.3	175	48.3
Bekele et al, January 2021[78]	Ethiopia	Cross-sectional online survey	Ethiopian 18 years or above	341	18−46	80.3	Fair	269	78.9	NR	NR	177	51.5

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								Event, n	%	Event, n	%	Event, n	%
Desalegn et al, January 2021[79]	Addis Ababa, Ethiopia	Community-based cross-sectional study	Adult aged >18 years	839	30.30±9.25	58.0	Fair	604	72.3	604	71.6	520	62.3
Habib et al, January 2021[80]	Kano, Nigeria	Cross-sectional community-based and online survey	general population including Health Care Workers (HCW) in Kano, Nigeria	886	28.58±10.25	55.4	Fair	576	65.3	629	71.4	576	65.0
Takoudjou Dzomo et al, January 2021[101]	N'Djamena, Chad	Cross-sectional study	individuals from N'Djamena who understood French and aged 18 years or older	2269	31.04±10.96	61.5	Poor	794	34.9	1384	61.4	688	30.3
Yoseph et al, January 2021[105]	Sidama, Ethiopia	Community-based cross-sectional study	Adult population who resided in the Sidama regional state for 6 months	1214	34±10	58.5	Fair	1093	90.0	1002	82.5	789	65.0
Mohamed et al, February 2021[85]	Sudan	Cross-sectional online survey	All Sudanese, aged 18 years and more	987	30.13±9.84	55.6	Good	839	85.1	819	83.0	849	86.0
Li et al, April 2021[83]	Lilongwe, Malawi	Cross-sectional community-based survey	Malawi residents at the age of 18 years or more living in Lilongwe	580	18−55+	35.0	Fair	278	48.0	371	64.0	232	40.0
Molla and Abegaz, April 2021[86]	Woldia town, Northeast Ethiopia	Community-based cross-sectional study	All the households that live in Woldia town who were aged 18 and above	404	18−39+	50.7	Fair	317	78.5	213	52.7	156	38.6
Taddese et al, April 2021[100]	Gondar, Ethiopia	Community-based cross-sectional study	People of age 18 years and above residing in Gondar city	623	33 ± 13.24	35.5	Poor	323	51.8	NR	NR	331	53.1

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									Event, n	%	Event, n	%	Event, n	%
Middle East	Al-Hanawi et al, May 2020[12]	Saudi Arabia	Cross-sectional online survey	aged 18 years or older living in KSA	3388	18−60+	41.9	Fair	2766	81.6	3252	95.9	2942	86.8
	Alahdal et al, June 2020[27]	Riyadh, Saudi Arabia	Cross-sectional online survey	18-65 years old	1767	18−65	25.0	Good	1100	62.3	1682	95.2	1406	79.6
	Honarvar et al, June 2020[47]	Shiraz, Iran	Cross-sectional and population-based study	people aged at least 15 years	1331	36.0	47.3	Good	679	50.9	818	61.5	481	36.2
	Naser et al, April 2020[59]	Jordan, Saudi Arabia and Kuwait	Cross-sectional online survey	aged 18 years who was living either in Jordan, Saudi Arabia or Kuwait	1208	18−50+	32.8	Fair	808	66.9	NR	NR	808	67.6
	Alhazmi et al, August 2020[28]	Saudi Arabia	Cross-sectional online survey	All citizens and residents over 18 years	1513	NR	45.0	Good	1230	81.3	1310	86.6	1239	81.9
	Domiati et al, August 2020[37]	Lebanon	Cross-sectional online survey	Residents of Lebanon	410	NR	41.9	Good	308	75.0	276	67.2	NR	NR
	Elayeh et al, October 2020[39]	Jordan	Cross-sectional online survey	Adult residents of Jordan	2104	18−55+	24.6	Good	1673	79.5	1236	58.7	1385	65.8
	Ghazi et al, September 2020[42]	Iraq	Cross-sectional online survey	Adults living in Iraq	272	36·4	58.1	Good	235	86.2	220	80.7	227	83.5
	Khaled et al, September 2020[53]	Aseer region, Saudi Arabia	Cross-sectional online survey	Residents of the Aseer Region, Saudi Arabia	740	18−70	11.5	Fair	594	80.3	533	71.9	575	77.6
	Zaid et al, July-August 2020[69]	Jordan	Cross-sectional online survey	Jordanian aged 18 years	3791	18−60	26.8	Poor	2114	55.7	NR	NR	NR	NR
	Nakhostin-Ansari et al, December 2020[88]	Iran	Cross-sectional online survey	Iranian at least 15 years old	1015	35.32±11.95	42.6	Fair	897	88.3	930	91.5	952	93.8
	Shahabi et al, January 2021[98]	Hormozgan, Iran	Cross-sectional online survey	Residents of Hormozgan Province aged above 15 years	2024	33.94 ± 9.37	35.6	Good	1607	79.4	1619	80.1	1864	92.1

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								Event, n	%	Event, n	%	Event, n	%
Tawalbeh et al, January 2021[102]	Jordan	Cross-sectional online survey	persons who aged 18 years or above, can read and write Arabic or English language	2470	18−40+	31.7	Poor	1976	80.3	2247	91.0	1929	78.1
Abbasi-Kangevari et al, February 2021[72]	Iran	Cross-sectional online survey	Iranian >18 years old	12232	32.2±9.9	38.1	Poor	9496	77.0	11481	93.1	10630	86.2
Al Ahdab et al, February 2021[73]	Syria	Cross-sectional online survey	Persons who were aged 16 years or more	706	16−50+	37.1	Good	417	59.0	433	61.3	521	73.8
Al-Hussami et al, February 2021[74]	Jordan	Cross-sectional online survey	Jordanian nationality aged 18 years or older	1076	34.83±10.50	40.2	Fair	904	84.4	926	86.0	849	78.8
Naqid et al, February 2021[89]	Kurdistan, Iraq	Community-based cross-sectional study	Kurdish population in Iraq aged >15 years	885	15−75	60.0	Poor	675	76.3	558	63.2	451	51.1
Saeed et al, February 2021[95]	Mosul, Iraq	Cross-sectional online survey	Mosul-Iraqi individuals above the age of 18 and currently residing in Mosul	909	18−50+	38.6	Fair	782	86.1	NR	NR	754	82.9
Omar and Amer, April 2021[90]	Egypt	Cross-sectional online survey	residents of Egypt using internet excluding the Illiterate, COVID-19 cases, and health-care provider	999	16−40+	33.3	Poor	808	80.9	794	79.5	529	52.7
Sakr et al, April 2021[96]	Lebanon	Cross-sectional online survey	Lebanese aged 18 years or above	1882	18−71	73.7	Fair	1263	67.1	1336	71.3	1412	75.0

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	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
North America	Alobuia et al, June 2020[29]	USA	Telephone survey	aged 18 year and older	2906	18−65+	48.0	Fair	2288	78.7	1502	51.7	1073	36.9
	Clements JM, May 2020[35]	USA	Cross-sectional online survey	US residents aged 18 years or older	1070	37.1	58.2	Good	838	78.3	NR	NR	536	50.1
	Czeisler et al, June 2020[36]	New York (NYC) and Los Angeles (LA), USA	Cross-sectional online survey	adults aged ≥18 years	1676	18−65+	43.9	Fair	NR	NR	USA (1286) NYC (242) LA (207)	USA (76.7) NYC (84.5) LA (80.0)	USA (920) NYC (178) LA (158)	USA (54.9) NYC (62.2) LA (61.0)
	Geldsetzer P, April 2020[41]	US and UK	Cross-sectional online survey	Adults registered with Prolific Academic Pty	US (2986) UK (2988)		US (49.1) UK (48.8)	Fair	US (2424) UK (2483)	US (81.2) UK (83.1)	NR	NR	NR	NR
	Leigh et al, October 2020[57]	Canada	Cross-sectional online survey	Adult Canadian	1996	18−65	45.2	Fair	1361	68.1	NR	NR	1143	57.2
South America	Bates et al, September 2020[34]	Ecuador	Cross-sectional online survey	Ecuadorians aged 18 or greater	2399	18−50+	37.0	Fair	1929	80.4	1331	55.5	2219	92.5
	Bates et al, November 2020[77]	Colombia	Cross-sectional online survey	People who identified as Colombians and who were aged 18 or greater	459	18−50+	28.1	Fair	353	76.8	289	63.0	424	92.3
	Bates et al, April 2021[76]	Venezuela	Cross-sectional online survey	Venezuelans and who were aged 18 or greater	3122	18−50+	24.6	Fair	2685	85.7	1592	50.6	2747	87.9
Europe	La Torre et al, June 2020[55]	Italy	Cross-sectional online survey	11-30 years	5234	11−30	36.7	Fair	3364	64.3	3707	70.8	NR	NR
	Mouchtouri et al, November 2020[87]	Greece	Telephone survey	Adult population of Greece	1858	49.2±17.4	41.2	Fair	1486	80.0	1319	71.0	1059	56.7

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	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
Oceania	Seal et al, June 2020[64]	Australia	Cross-sectional online survey	Australian adults (18 years and older)	1420	18−50+	48.0	Fair	816	57.4	NR	NR	803	56.5
Multiple continents	Masoud et al, February 2021[84]	Algeria, Brazil, Egypt, Ghana, India, Indonesia, Iraq, Ireland, Jordan, Lebanon, Libya, Morocco, Nigeria, Nepal, Palestine, Pakistan, Saudi Arabia, South Africa, Sri Lanka, Sudan, Syria, UK, USA	Cross-sectional online survey	Any citizen of the included 22 countries above the age of 18	71890	27.64±9.78	40.0	Good	49388	68.7	33429	46.5	62235	86.6

*NR-Not reported

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Table 6: Subgroup analysis by country income level, continent, study period, age group, gender, education and employment status

Outcome	Study characteristics	No of studies	Estimates score (%) (95% CI*), p-value	I^2	Egger test (p-value)
Knowledge	**Income level of the country**				
	Low income countries	16	72 (63−80), <0.001	99.36%	
	Lower-middle income countries	26	73 (68−78), <0.001	99.20%	
	Upper-middle income countries	25	79 (75−83), <0.001	99.49%	
	High income countries	15	72 (68−77), <0.001	99.03%	
	Continent				
	Asia	30	76 (71−81), <0.001	99.50%	
	Africa	21	71 (64−78), <0.001	99.33%	
	Middle East	20	75 (70−80), <0.001	99.19%	
	Americas	8	79 (75−83), <0.001	97.47%	
	Europe/Oceania	3	67 (54−80), <0.001	99.38%	
	Study period				
	January-March, 2020	28	73 (69−77), <0.001	99.52%	
	April-June, 2020	46	77 (74−80), <0.001	99.25%	
	July-October, 2020	8	67 (51-82), <0.001	99.72%	
	Age group	17			
	Below 30 years		78 (74−83), <0.001	97.86%	0.044
	30 years and above		80 (75−85), <0.001	98.93%	0.022
	Gender	46			
	Male		75 (71−79), <0.001	99.10%	0.051
	Female		74 (70−78), <0.001	99.39%	0.014
	Education level	39			
	Up to 12 years		71 (66−75), <0.001	98.41%	0.674
	Above 12 years		78 (74−82), <0.001	99.58%	0.081
	Employment status	20			
	Unemployed		74 (68−79), <0.001	97.30%	0.009
	Employed		77 (72−82), <0.001	99·48%	<0.001
	Retired/students		74 (68−81), <0.001	97.59%	0.632
Attitude	**Income level of the country**				
	Low income countries	12	69 (62−76), <0.001	98.73%	
	Lower-middle income countries	24	72 (66−78), <0.001	99.45%	
	Upper-middle income countries	23	78 (72−84), <0.001	99.68%	
	High income countries	12	77 (70−84), <0.001	99.51%	
	Continent				
	Asia	28	77 (72−82), <0.001	99.58%	
	Africa	17	70 (63−77), <0.001	99.15%	
	Middle East	17	79 (72−85), <0.001	99.58%	
	Americas	7	66 (56−77), <0.001	99.35%	
	Europe/Oceania	2	71 (70−72), 1.00	0.29%	
	Study period				
	January-March, 2020	20	80 (74−86), <0.001	99.71%	
	April-June, 2020	42	73 (68−77), <0.001	99.50%	
	July-October, 2020	8	65 (57-73), <0.001	98.70%	
	Age group	14			
	Below 30 years		68 (59−78), <0.001	99.45%	<0.001
	30 years and above		73 (65−81), <0.001	99.50%	0.010
	Gender	32			
	Male		73 (68−79), <0.001	99.55%	0.087
	Female		73 (68−78), <0.001	99.69%	0.004
	Education level	27			
	Up to 12 years		71 (66−76), <0.001	98.80%	0.451
	Above 12 years		71 (65−77), <0.001	99.71%	0.027
	Employment status	15			
	Unemployed		74 (66−81), <0.001	98.34%	0.029
	Employed		75 (69−82), <0.001	99.49%	0.006
	Retired/students		74 (66−82), <0.001	98.93%	0.004
Practice	**Income level of the country**				
	Low income countries	14	55 (46−64), <0.001	99.07%	
	Lower-middle income countries	24	74 (69−79), <0.001	99.36%	
	Upper-middle income countries	22	77 (70−84), <0.001	99.81%	
	High income countries	15	66 (58−73), <0.001	99.44%	
	Continent				
	Asia	29	76 (71−81), <0.001	99.62%	
	Africa	17	57 (49−65), <0.001	99.21%	
	Middle East	18	75 (68−82), <0.001	99.64%	
	Americas	9	66 (53−79), <0.001	99.72%	

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Outcome	Study characteristics	No of studies	Estimates score (%) (95% CI*), p-value	I^2	Egger test (p-value)
	Europe/Oceania	2	57 (55−59), 1.00	0.28%	
	Study period				
	January-March, 2020	23	67 (60−75), <0.001	99.79%	
	April-June, 2020	42	74 (70−79), <0.001	99.58%	
	July-October, 2020	8	57 (43−72), <0.001	99.65%	
	Age group	14			
	Below 30 years		80 (72−88), <0.001	99.34%	0.025
	30 years and above		82 (74−90), <0.001	99.70%	0.006
	Gender	39			
	Male		75 (69−80), <0.001	99.63%	<0.001
	Female		77 (72−82), <0.001	99.79%	<0.001
	Education level	31			
	Up to 12 years		74 (68−79), <0.001	99.24%	0.005
	Above 12 years		77 (71−83), <0.001	99.86%	<0.001
	Employment status	18			
	Unemployed		78 (72−85), <0.001	98.79%	<0.001
	Employed		76 (68−83), <0.001	99.70%	<0.001
	Retired/students		76 (68−84), <0.001	98.95%	0.174

*CI: Confidence Interval

Table 7: Analysis by some important questions regarding knowledge, attitude and practice on COVID-19

Outcome	Study characteristics	No of studies	Estimates score (%) (95% CI*), p-value	I^2	Egger test (p-value)
Knowledge	Q1.Main symptoms	61	85 (81−89), <0.001	99.76%	0.0001
	Q2.Population at risk	46	81 (77−86), <0.001	99.77%	<0.001
	Q3.Route of transmission	58	83 (79−87), <0.001	99.78%	<0.001
	Q4.Spread of infection by asymptomatic patients	39	75 (69−81), <0.001	99.80%	<0.001
	Q5.Face mask as a preventive measure	35	70 (63−78), <0.001	99.85%	0.699
	Q6.Avoidance of crowded places as a preventive measure	27	95 (93−97), <0.001	99.20%	0.030
	Q7.Isolation and treatment of patients to prevent infection	32	91 (87−95), <0.001	99.67%	0.012
	Q8.Quarantine/Incubation period	44	87 (82−91), <0.001	99.80%	0.041
Attitude	Q1.COVID-19 will be successfully controlled	22	71 (64−78), <0.001	99.78%	0.0003
	Q2.It is important to maintain social distance	11	88 (81−94), <0.001	99.25%	0.623
Practice	Q1.Hand hygiene	49	80 (75−86), <0.001	99.82%	0.016
	Q2.Wearing face mask	56	65 (58−73), <0.001	99.90%	0.003
	Q3.Social distancing	32	70 (63−77), <0.001	99.79%	0.0001
	Q4.Avoiding crowded places	45	75 (70−81), <0.001	99.77%	0.0001

*CI: Confidence Interval

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Figure 1: Correlation between components of KAP

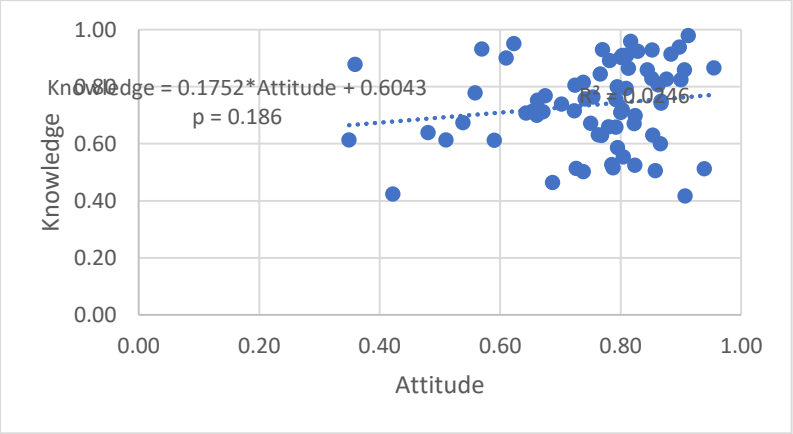


Figure 1a: Correlation between knowledge and attitude

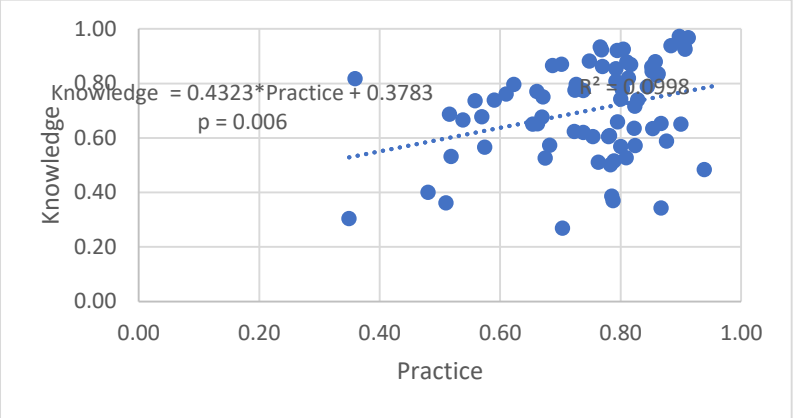


Figure 1b: Correlation between knowledge and practice

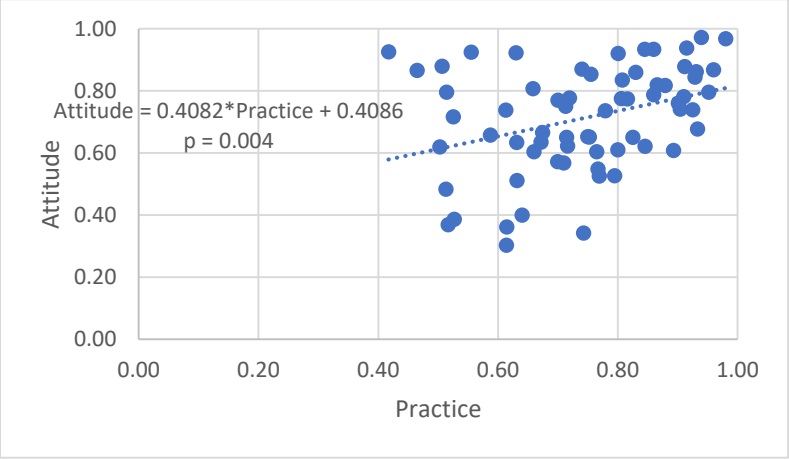
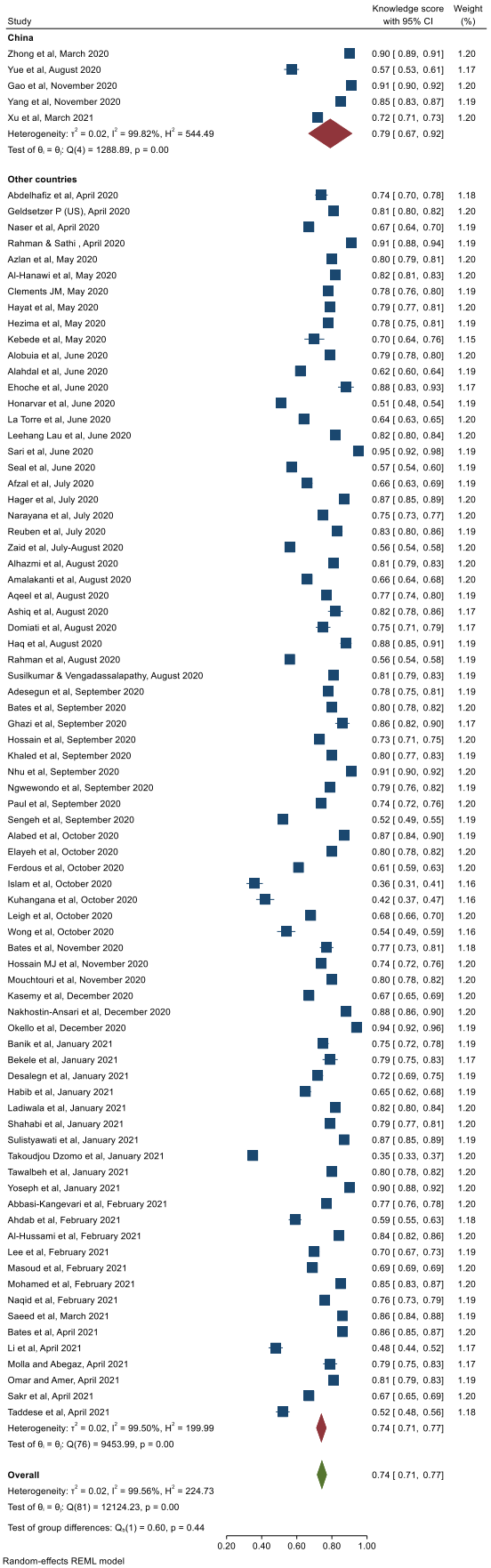


Figure 1c: Correlation between attitude and practice

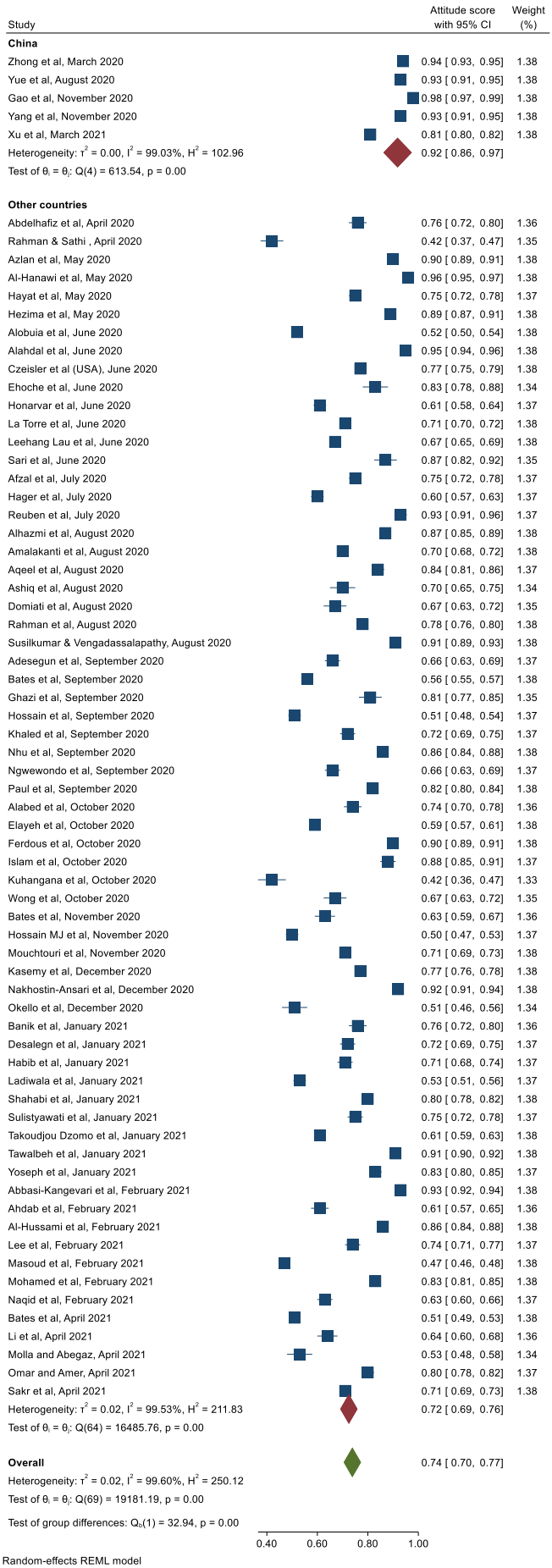
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Figure 2: China vs Other countries

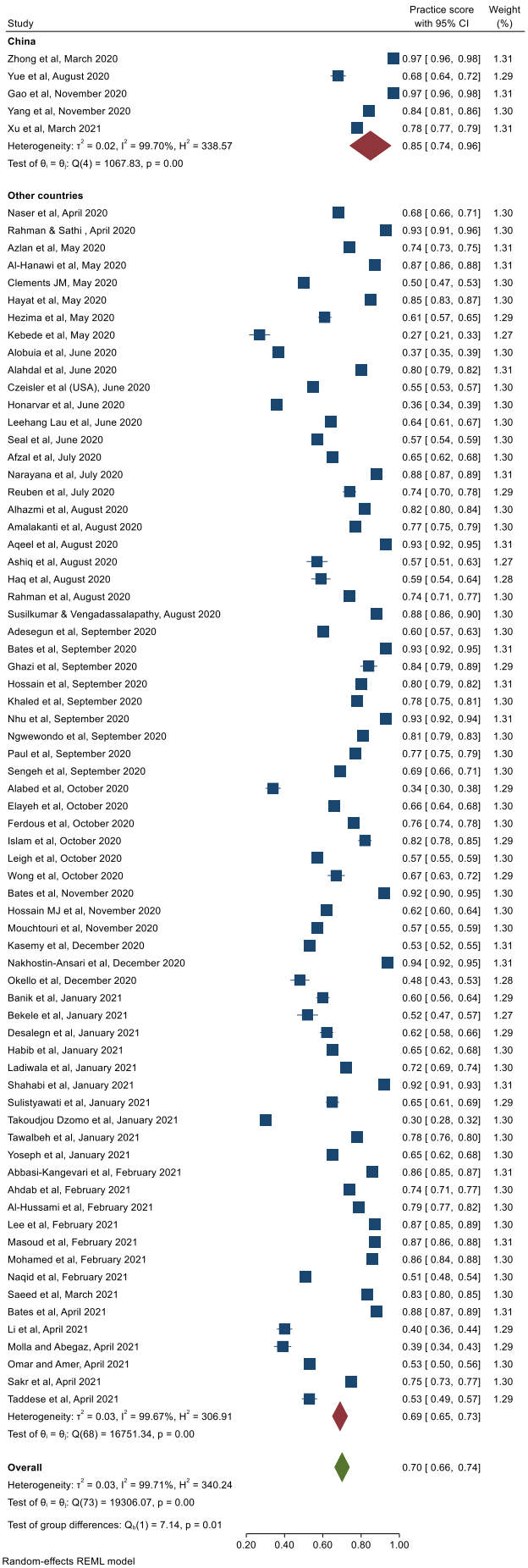


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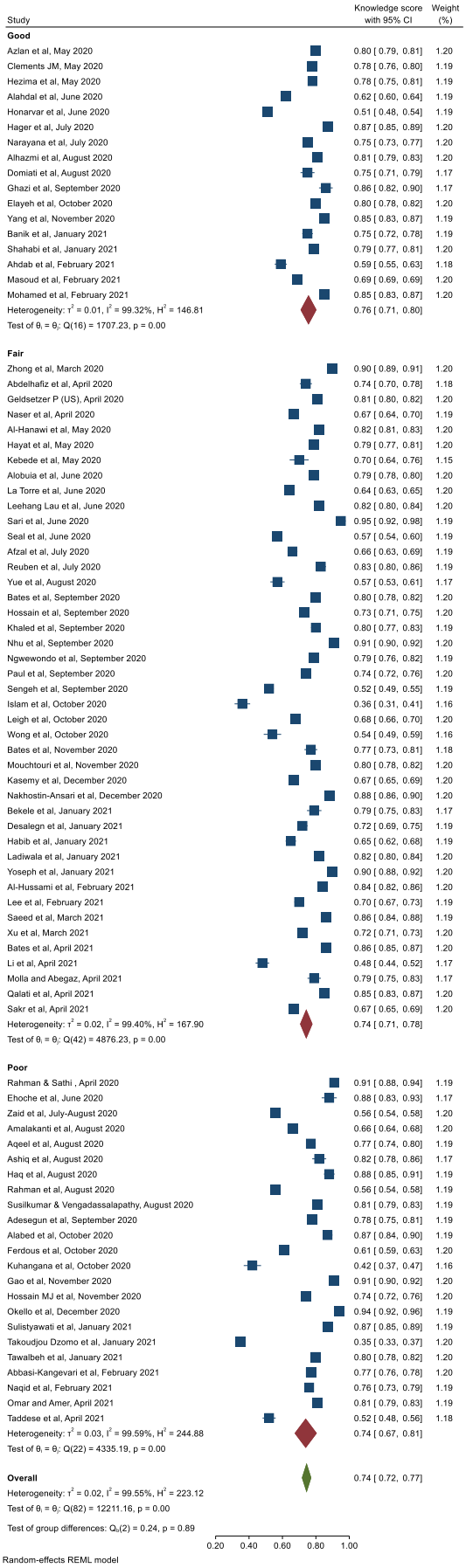


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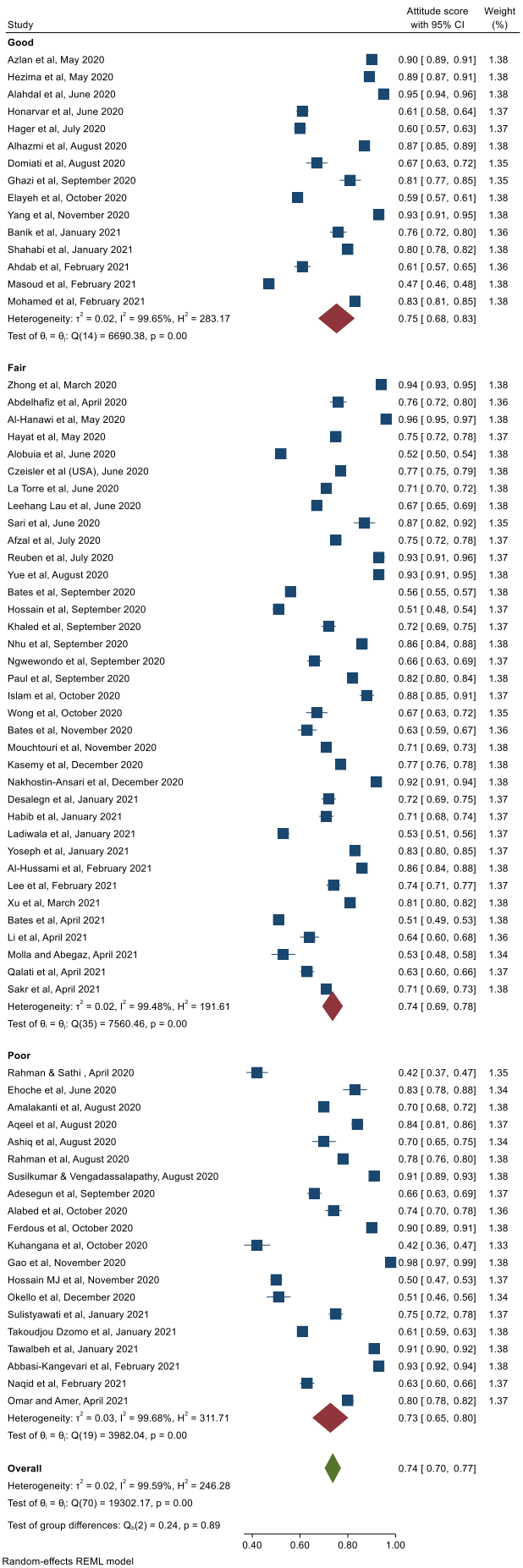


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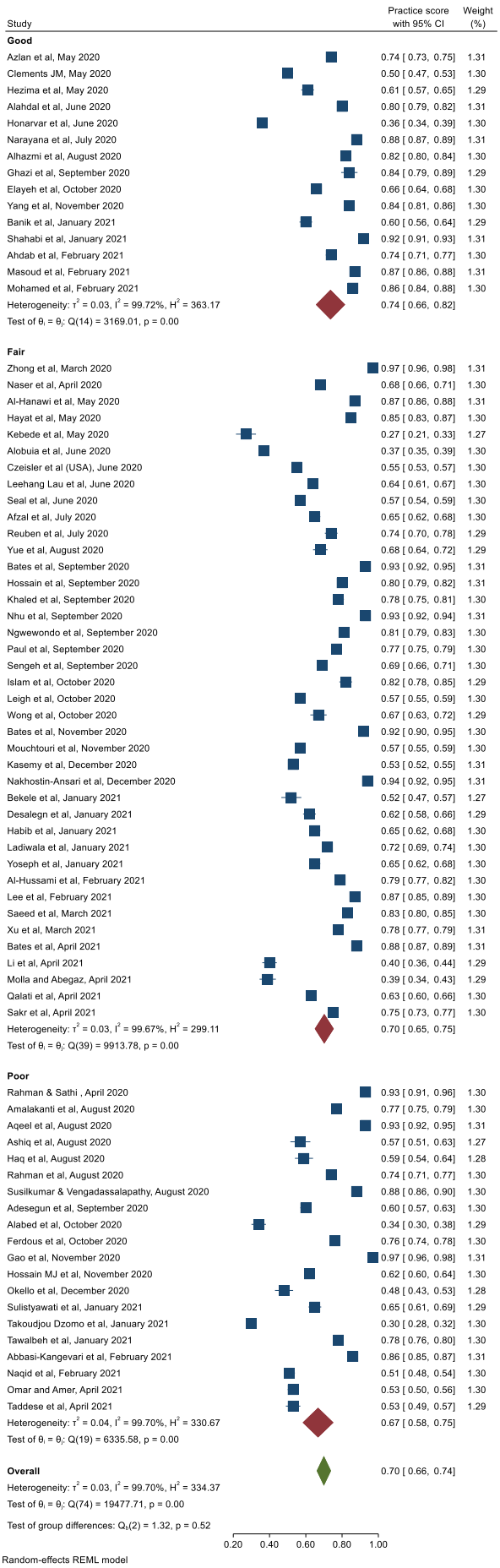
Figure 3: Analysis by quality of the studies



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